

NILPOTENT LIE GROUPS AND SUBLINEAR BILIPSCHITZ EQUIVALENCE

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ABSTRACT. As it turns out, the quasiisometric classification of compactly generated locally compact groups of polynomial growth reduces to that of simply connected nilpotent Lie groups. The large-scale geometry of these groups exhibits some sublinear features that we will explore in these lectures. In particular, we will see how sublinear bilipschitz equivalences (generalized quasiisometries) arise between certain pairs of simply connected nilpotent Lie groups, and reframe the quasiisometric classification problem into a quantitative perspective.

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This minicourse is about certain sublinear aspects in the large-scale geometry of nilpotent groups. Our ultimate aim is to explain a simple version of [Cor19, Section 6] and the questions it raises. We only assume knowledge of a few terms in geometric group theory and made an effort to be otherwise self-contained, although we will admit some results. Some contents need more background to be fully appreciated; they are marked with an asterisk (*). Parts of the material we present here have been exposed, with a wider scope, in the following two resources:

- Pierre Pansu’s June 2023 minicourse in Pisa on the large-scale geometry of nilpotent groups, which was recorded and is available here : [Lectures 1 and 2](#) (starting at 0:55:21) and [Lecture 3](#) (starting at 2:49:05). The slides are [available on Pansu’s website](#).
- Chapters 8 to 11 in Enrico Le Donne’s textbook [LD25] (with more prerequisites in differential and metric geometry than here, some of it being covered in the

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earlier chapters). A fairly complete version of this book is available [on the arXiv](#). There are many exercises.

In addition, the minicourse likely has some overlaps with Yves Cornulier’s course in the 2019 Jyväskylä summer school and with the minicourse by Claudio Llosa Isenrich and Romain Tessera in the “Équivalence orbitale” meeting at Besse-en-Chandesse in 2024.

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0. MOTIVATION

Let G be a compactly generated locally compact (thereafter CGLC) group. We say that G has **polynomial growth** if for some (any) compact generating set S of G , $\text{vol}(S^n) \leq \text{poly}(n)$, where vol is some (any) Haar measure on G . Being of polynomial growth is a quasiisometry (QI) invariant of CGLC groups.

Theorem 0.1 (... Losert¹ [Los20, Theorems 2 and 3]). *Let G be a CGLC group of polynomial growth. Then*

- (1) G has a maximal compact normal subgroup $W(G)$.
- (2) $G/W(G)$ embeds with closed and co-compact image into a group of the form $N \rtimes Q$, where N is a **simply connected nilpotent Lie group**, and Q is a compact group.

Moreover, given G as above, N is the unique simply connected nilpotent Lie group such that (2) holds.

(We will explain what a simply connected nilpotent Lie group is, but you can take it as a black-box now.) Since $G/W(G)$ and N both sit as closed co-compact subgroups in $N \rtimes Q$, the groups G and N are quasiisometric. We call N the **connected nilshadow** of G . Here is a long-standing open problem :

Problem : Classify CGLC groups of polynomial growth up to quasiisometry.

An early version was stated in [Pan89] where the problem is attributed to Gromov. More recently the following conjecture has been made. A slightly weaker form of the conjecture was formulated in [Lüc08], and appears as a question in [Ghy90, p.208] and [FM00].

Conjecture 0.2 (Cornulier, [Cor18]). *N and N' are simply connected nilpotent Lie groups. If N and N' are quasiisometric, then N and N' are isomorphic as Lie groups.*

Conjecture 0.2 would settle the QI classification problem in the following way: Theorem 0.1 associates uniquely two simply connected nilpotent Lie groups N and N' to CGLC groups G and G' ; if G and G' are QI, then N and N' are QI as well. According to the conjecture, they should be isomorphic. (Conversely, if $N \simeq N'$ then G and G' are QI). Thus, tentatively, G and G' should be QI if and only if their connected nilshadows are isomorphic as Lie groups.

¹Historical comments: a) if you replace “polynomial growth” by “nilpotent” in the statement of Theorem 0.1, this is easier and was already known to Guivarc’h in 1973; the general case incorporates Gromov’s well-known theorem on groups with polynomial growth. b) (1) is already [Los01, Prop 1]. An equivalent statement appears in [Cor15, Remark 2.11].

1. NILPOTENT LIE ALGEBRAS

Let V be a finite-dimensional real vector space. A **Lie bracket** on V is a bilinear map from $V \times V$ to V , denoted

$$(X, Y) \rightarrow [X, Y]$$

such that $[Y, X] = -[X, Y]$ and

$$(J) \quad [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in V$. (Note that antisymmetry implies that $[X, \lambda X] = [\lambda X, X] = 0$ for all $\lambda \in \mathbf{R}$.) A **Lie algebra** is a pair $\mathfrak{g} = (V, [\cdot, \cdot])$. A **homomorphism of Lie algebras** is a linear homomorphism which sends the Lie bracket of a pair to the Lie bracket of their images.

Example 1.1. For all $n \geq 1$, $\mathfrak{gl}_n(\mathbf{R}) := (\text{End}(\mathbf{R}^n), (X, Y) \mapsto XY - YX)$ is a Lie algebra. (J) is a consequence of the associativity in $\text{End}(\mathbf{R}^n)$.

If X is an element of a Lie algebra $\mathfrak{g}$, we let $\text{ad}_X = [X, -]$ and call $\text{ad}_\cdot : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$ the **adjoint representation**. The next exercise explains the terminology.

Exercise A. Let $\mathfrak{g}$ be a Lie algebra. Using (J), check that $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a homomorphism of Lie algebras.

A **Lie monomial** over an alphabet of letters $\{x_i\}$ is a formal expression of iterated brackets of letters, e.g. $\mu = [x_1, [x_2, [[x_3, x_4], x_5]]]$ is a Lie monomial over $\{x_1, \dots, x_5\}$; a **simple Lie monomial** is a Lie monomial where the iteration on brackets is always made to the left, e.g. μ above is not a simple Lie monomial, while

$$\sigma = [x_1, [x_2, [x_1, [x_3, x_2]]]]$$

is one. To save space we will abbreviate simple Lie monomials and their evaluations by omitting brackets, e.g. we will write $\sigma = [x_1, x_2, x_1, x_3, x_2]$. The **length** of a Lie monomial is the number of letters appearing in it, e.g. μ and σ have length 5. Using (J) and antisymmetry one may prove that every Lie monomial is formally a linear combination of simple Lie monomials of the same length with coefficients in $\{\pm 1\}$. (By “formally” we mean that the situation is exactly analogous to that of polynomial identities holding in any $\mathbf{R}$ -algebra, or even actually in any ring.)

Definition 1.2 (Lie subalgebra and Lie span). Let $\mathfrak{g} = (V, [\cdot, \cdot])$ be a Lie algebra. A **Lie subalgebra** of $\mathfrak{g}$ is a subspace $W \subset V$ stable under $[\cdot, \cdot]$, that we equip with the restriction of the bracket. Given some subset $S \subseteq \mathfrak{g}$, we denote by $\text{Liespan}(S)$ the smallest Lie subalgebra of $\mathfrak{g}$ containing S and call it the **Lie subalgebra generated by S** . $\text{Liespan}(S)$ is the set of formal sums of (simple) Lie monomials with real coefficients, in which we substitute elements of S .

A Lie subalgebra $\mathfrak{w}$ of a Lie algebra $\mathfrak{g}$ is an **ideal** if

$$[\mathfrak{g}, \mathfrak{w}] \subseteq \mathfrak{w}$$

where $[\mathfrak{g}, \mathfrak{w}]$ denotes $\text{span}\{[X, Y] : X \in \mathfrak{g}, Y \in \mathfrak{w}\}$. In this case $\mathfrak{g}/\mathfrak{w}$ is endowed with the structure of a Lie algebra, by letting $[X + \mathfrak{w}, Y + \mathfrak{w}] := [X, Y] + \mathfrak{w}$. Ideals are exactly the kernels of Lie algebras homomorphisms. An example of ideal is the **center** $Z(\mathfrak{g}) = \ker \text{ad}$. The following general construction yields a series of nested ideals.

Definition 1.3. Let $\mathfrak{g}$ be a Lie algebra. Define its **descending central series** $(C^k \mathfrak{g})_{k \geq 1}$ as follows. $C^1 \mathfrak{g} = \mathfrak{g}$ and, for $k \geq 1$, $C^{k+1} \mathfrak{g} = [\mathfrak{g}, C^k \mathfrak{g}]$.

Lemma 1.4. For all $k, \ell \geq 1$, $[C^k \mathfrak{g}, C^\ell \mathfrak{g}] \subseteq C^{k+\ell} \mathfrak{g}$.

Proof. By induction on ℓ . For $\ell = 1$, $[C^k \mathfrak{g}, C^\ell \mathfrak{g}] = [\mathfrak{g}, C^k \mathfrak{g}] = C^{k+1} \mathfrak{g} \subseteq C^{k+1} \mathfrak{g}$. Assume it's true for some $\ell \geq 1$. Then

$$\begin{aligned} [C^k \mathfrak{g}, C^{\ell+1} \mathfrak{g}] &= [C^k \mathfrak{g}, \mathfrak{g}, C^\ell \mathfrak{g}] \\ &\stackrel{(J)}{\subseteq} [\mathfrak{g}, C^\ell \mathfrak{g}, C^k \mathfrak{g}] + [C^\ell \mathfrak{g}, C^k \mathfrak{g}, \mathfrak{g}] \\ &\subseteq [\mathfrak{g}, C^{k+\ell} \mathfrak{g}] + C^{k+1+\ell} \mathfrak{g} \quad (\text{by the induction hypothesis}) \\ &= C^{k+\ell+1} \mathfrak{g}. \end{aligned} \quad \square$$

Definition 1.5. We say that $\mathfrak{g}$ is **nilpotent** if there exists $s \in \mathbf{N}$ such that $C^{s+1} \mathfrak{g} = 0$. The minimal such s is called the **nilpotency class** of $\mathfrak{g}$; we also say that $\mathfrak{g}$ is s -nilpotent.

For a Lie algebra, being nilpotent passes to subalgebras and quotients. An ordered basis $(e_1, \dots, e_d)$ of a nilpotent Lie algebra $\mathfrak{g}$ is **adapted** if for some sequence (i_k) we have that for all $k \geq 1$, $C^k \mathfrak{g} = \text{span}(e_{i_k}, \dots, e_d)$. Every nilpotent Lie algebra has an adapted basis.

Example 1.6. Let $n \geq 2$ and define the following Lie subalgebra of $\mathfrak{gl}_n(\mathbf{R})$:

$$\mathfrak{u}_n = \text{span}\{E_{ij} : j > i\} \subseteq \mathfrak{gl}_n(\mathbf{R}),$$

where E_{ij} denotes $e_j^* \otimes e_i$, $e_1, \dots, e_n$ being the canonical basis of $\mathbf{R}^n$. Let us check that $\mathfrak{u}_n$ is $(n-1)$ -nilpotent and find an adapted basis. Letting $E_{i_1 j_1}$ and $E_{i_2 j_2}$ in $\mathfrak{u}_n$, we have

$$[E_{i_1 j_1}, E_{i_2 j_2}] = \begin{cases} E_{i_1 j_2} & i_2 = j_1 \\ -E_{i_2 j_1} & j_2 = i_1 \\ 0 & \text{otherwise,} \end{cases}$$

and if $i_2 = j_1$ then $j_2 > i_2 = j_1 > i_1$, so $j_2 > i_1$ and $E_{i_1 j_2} \in \mathfrak{u}_n$; the same argument shows that $E_{i_2 j_1} \in \mathfrak{u}_n$. This proves that $\mathfrak{u}_n$ is indeed a Lie algebra. Now note that if $j_2 \geq i_2 + k$ and $i_2 = j_1$ then $j_2 \geq j_1 + k \geq i_1 + k + 1$; equality can occur if $j_1 = i_1 + 1$, and similarly. This proves that

$$C^k \mathfrak{u}_n = \text{span}\{E_{ij} : j \geq i + k\} = \left\{ \begin{pmatrix} 0 & \cdots & 0 & x_{1,k+1} & \cdots & x_{1,n} \\ \vdots & & & \ddots & \ddots & \vdots \\ \vdots & & & & \ddots & x_{k+1,1} \\ \vdots & & (0) & & & 0 \\ \vdots & & & & & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 \end{pmatrix} : x_{i,j} \in \mathbf{R} \right\}.$$

Especially, $C^{n-1} \mathfrak{u}_n = \text{span}(E_{1,n}) \neq 0$ and $C^n \mathfrak{u}_n = 0$, so $\mathfrak{u}_n$ is $(n-1)$ -nilpotent, and an adapted basis is

$$(E_{1,2}, E_{2,3}, \dots, E_{n-1,n}, E_{1,3}, \dots, E_{1,n-1}, E_{2,n}, E_{1,n}).$$

Proposition and Definition 1.7. Let $\mathfrak{g}$ be a s -nilpotent Lie algebra with $s \geq 2$. Then $\dim \mathfrak{g} \geq s + 1$. Equality can occur; in this case we call $\mathfrak{g}$ **filiform**.

Proof. Since $C^{s+1}\mathfrak{g} = 0$ we have

$$(1.1) \quad \dim \mathfrak{g} = \sum_{k=1}^s \dim(C^k \mathfrak{g} / C^{k+1} \mathfrak{g}).$$

Assume towards contradiction that there exists $k \in \{1, \dots, s\}$ such that $C^k \mathfrak{g} = C^{k+1} \mathfrak{g}$, then $C^\ell \mathfrak{g} = C^k \mathfrak{g}$ for all $\ell > k$. Since $\mathfrak{g}$ is nilpotent we have $C^k \mathfrak{g} = 0$. But this contradicts the minimality of s . So every term in the sum (1.1) is at least equal to 1, and $\dim \mathfrak{g} \geq s$.

We now prove that $\dim(\mathfrak{g}/C^2 \mathfrak{g}) > 1$. We know that $s \geq 2$ so there exists some $Z \in C^2 \mathfrak{g} \setminus C^3 \mathfrak{g}$. We can write

$$(1.2) \quad Z = [X_1, Y_1] + \dots + [X_r, Y_r]$$

We want to show that there exists $i \in \{1, \dots, r\}$ such that $X_i + C^2 \mathfrak{g}$ and $Y_i + C^2 \mathfrak{g}$ are linearly independent. Otherwise, assuming as we may that the X_i s are nonzero, $Y_i \in \lambda_i X_i + C^2 \mathfrak{g}$ for some $\lambda_i \in \mathbf{R}$. But then $[X_i, Y_i] \in [X_i, \lambda_i X_i] + [\mathfrak{g}, C^2 \mathfrak{g}] \subseteq C^3 \mathfrak{g}$. This is a contradiction, as $Z \notin C^3 \mathfrak{g}$.

We now show that the equality $\dim \mathfrak{g} = s + 1$ can occur. Equip $V = \mathbf{R}^{s+1} = \text{span}(X_1, \dots, X_{s+1})$ with the Lie bracket $[X_1, X_i] = X_{i+1}$ for $i = 2, \dots, s$, and $[X_k, X_j] = 0$ otherwise. We check (J). If $\min(k, j) > 1$ then

$$[X_1, [X_k, X_j]] + [X_k, [X_j, X_1]] + [X_j, [X_1, X_k]] = [X_k, X_{j+1}] + [X_j, X_{k+1}] = 0$$

(with the convention $X_{s+2} = 0$). And for every $j > 1$

$$[X_1, [X_1, X_j]] + [X_1, [X_j, X_1]] + [X_j, [X_1, X_1]] = X_{j+2} - X_{j+2} = 0$$

(again with the convention $X_{s+2} = X_{s+3} = 0$). If j, k, ℓ are all greater than 1, then any bracket of X_j, X_k and X_ℓ in any order is zero. So (J) holds indeed. $\square$

Definition 1.8. We call the Lie algebra we constructed in the proof $\mathfrak{l}_{s+1}$ and call it the **model filiform** Lie algebra of class s .

The Lie algebra $\mathfrak{u}_3$ is isomorphic to $\mathfrak{l}_3$. We call it the **Heisenberg** Lie algebra and denote it by $\mathfrak{heis}$. This is the prototype of a non-abelian nilpotent Lie algebra.

Exercise B. (1) Show that every non-abelian nilpotent Lie algebra of dimension three is isomorphic to $\mathfrak{heis}$.

(2) Show that the 4-dimensional nilpotent Lie algebras are isomorphic either to $\mathbf{R}^4$ (by which we mean the Lie algebra $\mathbf{R}^4$ with the Lie bracket 0), $\mathfrak{heis} \times \mathbf{R}$ or $\mathfrak{l}_4$, and that the latter three Lie algebras are pairwise non-isomorphic.

There is a convenient way to denote the Lie brackets in a basis of a Lie algebra. See Figure 1. In theory things can get more complicated, but these diagrams are sufficient to denote all nilpotent Lie algebras of dimension less or equal 6.

Exercise C. Let $\mathfrak{h}$ be a 2-nilpotent Lie algebra such that $\dim(Z(\mathfrak{h})) = 1$.

- (1) Show that $\dim \mathfrak{h}$ is odd; we will let n be such that $\dim \mathfrak{h} = 2n + 1$.
- (2) Show that $\mathfrak{h}$ has a basis $(X_1, \dots, X_n, Y_1, \dots, Y_n, Z)$ such that $Z \in Z(\mathfrak{g})$,

$$[X_i, Y_j] = \begin{cases} Z & i = j \\ 0 & \text{otherwise;} \end{cases}$$

draw the diagram representing $\mathfrak{h}$ in this basis.

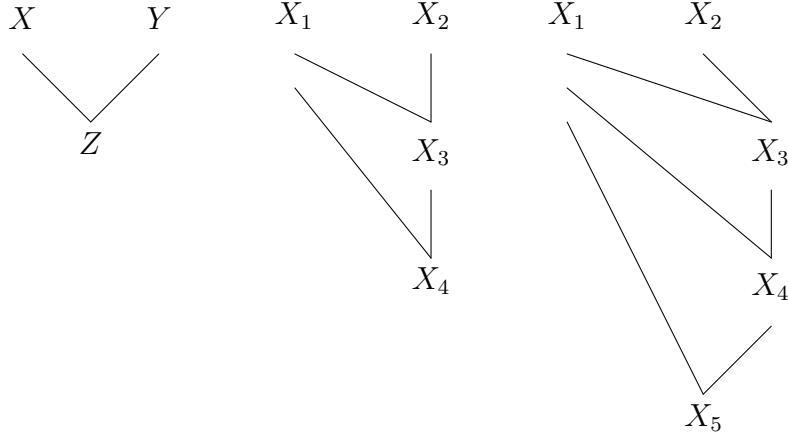


FIGURE 1. Depiction of the defining brackets of model filiform Lie algebras $\mathfrak{heis} = \mathfrak{l}_3, \mathfrak{l}_4$ and $\mathfrak{l}_5$ in adapted bases.

- (3) Show that $\mathfrak{h}$ is isomorphic to a Lie subalgebra of $\mathfrak{u}_{n+2}$. Is it isomorphic to a Lie subalgebra of $\mathfrak{u}_{n+1}$?

We call these Lie algebras the (higher-dimensional) Heisenberg Lie algebras.

2. CARNOT LIE ALGEBRAS

Proposition and Definition 2.1 (Carnot algebra). *Let $\mathfrak{g}$ be a s -nilpotent Lie algebra. We define a new Lie algebra, $\mathfrak{g}_\infty$, that we will call the **associated Carnot algebra** of $\mathfrak{g}$, in the following way. As a vector space,*

$$\mathfrak{g}_\infty = \bigoplus_{k=1}^s C^k \mathfrak{g} / C^{k+1} \mathfrak{g}.$$

For $X \in C^k \mathfrak{g}$ and $Y \in C^\ell \mathfrak{g}$, we define $\{X + C^{k+1} \mathfrak{g}, Y + C^{\ell+1} \mathfrak{g}\} := [X, Y] + C^{k+\ell+1} \mathfrak{g}$. After extending by linearity, $(X, Y) \mapsto \{X, Y\}$ defines a Lie bracket on $\mathfrak{g}_\infty$, with the following properties :

C1. $\mathfrak{g}_\infty$ is s -nilpotent.

C2. Denote $(\mathfrak{g}_\infty)_k := C^k \mathfrak{g} / C^{k+1} \mathfrak{g}$. Then $((\mathfrak{g}_\infty)_k)_k$ defines a **grading of $\mathfrak{g}_\infty$ by the positive integers**: $\mathfrak{g}_\infty = (\mathfrak{g}_\infty)_1 \oplus \cdots \oplus (\mathfrak{g}_\infty)_s$ and $[(\mathfrak{g}_\infty)_k, (\mathfrak{g}_\infty)_\ell] \subseteq (\mathfrak{g}_\infty)_{k+\ell}$.

C3. $\mathfrak{g}_\infty = \text{Liespan}((\mathfrak{g}_\infty)_1)$.

If $\mathfrak{g} \simeq \mathfrak{g}_\infty$ then we say that $\mathfrak{g}$ is itself a **Carnot Lie algebra**. A direct sum decomposition of a Lie algebra is called a **Carnot grading** if it has the properties **C2** and **C3** above; a Lie algebra is Carnot if and only if it possesses a Carnot grading.

Sketch of proof. We prove that the Lie bracket is well defined in $\mathfrak{g}_\infty$. Given $X \in C^k \mathfrak{g}$ and $Y \in C^\ell \mathfrak{g}$, let $X' \in X + C^{k+1} \mathfrak{g}$ and $Y' \in Y + C^{\ell+1} \mathfrak{g}$. Then

$$[X', Y'] - [X, Y] \in [C^k \mathfrak{g}, C^{\ell+1} \mathfrak{g}] + [C^{k+1} \mathfrak{g}, C^\ell \mathfrak{g}] + [C^{k+1} \mathfrak{g}, C^{\ell+1} \mathfrak{g}] \subseteq C^{k+\ell+1} \mathfrak{g}$$

by Lemma 1.4. This shows that the Lie bracket is well defined in $\mathfrak{g}_\infty$; linearity, antisymmetry, and (J) follow from that of the Lie bracket in $\mathfrak{g}$. Property C2 follows again from Lemma 1.4. By induction on $k \geq 1$, we prove that

$$C^k \mathfrak{g}_\infty = \bigoplus_{i \geq k} (\mathfrak{g}_\infty)_i$$

and that $[(\mathfrak{g}_\infty)_1, (\mathfrak{g}_\infty)_k] = (\mathfrak{g}_\infty)_{k+1}$. $\square$

The terminology of Carnot Lie algebra is the most used in geometric group theory; it was introduced by Pansu in a paper concerned with QI rigidity [Pan89]. However, authors from the PDE community (and sometimes also in subriemannian geometry) have studied these Lie algebras since longer and rather call them **stratified Lie algebras**. In relation to certain problems of prolongations of geometric structures, some grade them with *negative* integers [Tan70] (See a minicourse by Alessandro Ottazzi). The process of going from $\mathfrak{g}$ to $\mathfrak{g}_\infty$ is an example of what physicists have called a **Lie algebra contraction** and it plays a role when taking certain limits (either from relativistic to classical, or from quantum to classical physics) [IW53, MR206]. One slogan is that the contracted Lie algebra is “more abelian” than the one we started with; for instance, the dimension of the center increases. We work out a few examples below before discussing a few subtleties.

Example 2.2. The Heisenberg Lie algebra, and more generally the nilpotent Lie algebras $\mathfrak{u}_n$ and $\mathfrak{l}_n$ are Carnot. This can be checked by making the corresponding gradings explicit; in fact we already have adapted bases (in the case of $\mathfrak{u}_n$ this was done in Example 1.6), the gradings can be obtained by concatenating linear spans of subsequences of elements of from these bases.

Example 2.3. Consider the Lie algebra

$$\mathfrak{g}_{4,3} = \text{span}(X_1, X_2, X_3, Y_1, Y_2, Z)$$

where $[X_1, X_2] = X_3$, $[X_1, X_3] = [Y_1, Y_2] = Z$, and all other brackets are zero. Some computations give that $C^2\mathfrak{g}_{4,3} = \text{span}(X_3, Z)$, that $C^3\mathfrak{g}_{4,3} = \text{span}(Z)$, and that $\mathfrak{g}_{4,3}$ is 3-nilpotent. Moreover, the basis $(X_1, X_2, Y_1, Y_2, X_3, Z)$ is adapted, and

$$\{Y_1 + C^2\mathfrak{g}_{4,3}, Y_2 + C^2\mathfrak{g}_{4,3}\} = Z + C^3\mathfrak{g}_{4,3} = 0$$

in $C^2\mathfrak{g}_{4,3}/C^3\mathfrak{g}_{4,3}$. It follows that $(\mathfrak{g}_{4,3})_\infty \simeq \mathfrak{l}_4 \times \mathbf{R}^2$ where the $\mathfrak{l}_4$ factor is generated by $X_1, X_2 \bmod C^2\mathfrak{g}_{4,3}$, $X_3 \bmod C^3\mathfrak{g}_{4,3}$ and Z . A little of linear algebra shows that $Z(\mathfrak{g}_{4,3}) = \text{span}(Z)$ is 1-dimensional, while $Z((\mathfrak{g}_{4,3})_\infty)$ is 3-dimensional. So $\mathfrak{g}_{4,3}$ is not Carnot.

Exercise D. Consider the Lie algebra

$$\mathfrak{l}'_5 = \text{span}(X_1, X_2, X_3, X_4, X_5)$$

where $[X_1, X_2] = X_3$, $[X_1, X_3] = X_4$ and $[X_1, X_4] = [X_2, X_3] = X_5$.

- (1) Compute the lower central series of $\mathfrak{l}'_5$ and show that $(\mathfrak{l}'_5)_\infty \simeq \mathfrak{l}_5$.
- (2) Show that $\mathfrak{l}_5$ and $\mathfrak{l}'_5$ are not isomorphic Lie algebras. (Hint: you may look at ranks of ad_X for various X in $\mathfrak{l}_5$ and $\mathfrak{l}'_5$.)

Exercise E. (1) Show that every 2-nilpotent Lie algebra is Carnot.

- (2) Let $\mathfrak{g}$ be a s -nilpotent Lie algebra and let

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s = \mathfrak{g}'_1 \oplus \cdots \oplus \mathfrak{g}'_s$$

be two Carnot gradings on $\mathfrak{g}$. Show that there exists a Lie algebra automorphism of $\mathfrak{g}$ sending $\mathfrak{g}_i$ onto $\mathfrak{g}'_i$ for all $i = 1, \dots, s$.

Beware that it is in general not true that in a Carnot Lie algebra, any supplementary subspace to $[\mathfrak{g}, \mathfrak{g}]$ is the $\mathfrak{g}_1$ of a Carnot grading $\mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s$. See [Cor16, Remark 3.16] for a counterexample.

Proposition and Definition 2.4. *Let $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s$ be a Carnot Lie algebra. Decompose every $X \in \mathfrak{g}$ into $X_1 + \cdots + X_s$ where $X_i \in \mathfrak{g}_i$ for all i . For all $t \in \mathbf{R}_{>0}$, define*

$$t \bullet X := tX_1 + \cdots + t^s X_s.$$

*Then $t \bullet -$ is a Lie algebra automorphism of $\mathfrak{g}$, called a **Carnot dilation**.*

Proof. Let $X \in \mathfrak{g}_k$ and $Y \in \mathfrak{g}_\ell$. Then $[X, Y] \in \mathfrak{g}_{k+\ell}$, so

$$t \bullet [X, Y] = t^{k+\ell} [X, Y] = [t^k X, t^\ell Y] = [t \bullet X, t \bullet Y].$$

We may now conclude that the same identity holds for every $X, Y \in \mathfrak{g}$ by linearity of the bracket. $\square$

It is worth noting, at this point, that in $\mathbf{R}^d$, the fact that Lie algebra automorphisms (which are just elements of $\mathfrak{gl}_d(\mathbf{R})$) commute with the dilation (which is just the multiplication by scalar t) is a consequence of their mere linearity, but in a non-abelian Carnot Lie algebra, it's not. We say that a homomorphism $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$ between Carnot Lie algebras is a **Carnot homomorphism** if it intertwines the Carnot dilations of $\mathfrak{g}$ and $\mathfrak{h}$, namely, for all $X \in \mathfrak{g}$ and $t \in \mathbf{R}_{>0}$, $t \bullet \psi(X) = \psi(t \bullet X)$.

Although $\mathfrak{g}$ and $\mathfrak{g}_\infty$ are isomorphic as vector spaces, there is no canonical linear isomorphism $L: \mathfrak{g} \rightarrow \mathfrak{g}_\infty$, and indeed no canonical linear isomorphism respecting the lower central series.

3. NILPOTENT LIE GROUPS

Theorem 3.1 (Consequence of Ado's theorem). *Let $\mathfrak{g}$ be a nilpotent Lie algebra. Then $\mathfrak{g}$ is isomorphic to a Lie subalgebra of $\mathfrak{u}_n$ for large enough n .*

Ado's theorem is quite deep, see e.g. Bourbaki [Bou82, Chapter 1, §7] for a proof.

Exercise F. The aim of this exercise is to prove Theorem 3.1 without resorting to Ado's theorem in the special case when $\mathfrak{g}$ is Carnot.

- (1) Let $\mathfrak{h} = \mathfrak{g} \oplus \langle T \rangle$ where $[T, X] = iX$ for all $X \in \mathfrak{g}_i$. Show that $\mathfrak{h}$ is a Lie algebra and that $\{X \in \mathfrak{h} : \text{ad}_X = 0\} = 0$ where $\text{ad}_X(Y)$ denotes $[X, Y]$.
- (2) Show that $\text{ad}: \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{h})$, $X \mapsto \text{ad}_X$ defines an injective Lie algebra homomorphism, i.e. $[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]}$. Deduce from there that $\mathfrak{g}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}_{\dim \mathfrak{g}+1}(\mathbf{R})$.
- (3) Let $n = \dim \mathfrak{g}$. Show that there is a basis $(e_1, \dots, e_{n+1})$ of $\mathfrak{h}$ such that $\text{ad}(\mathfrak{g})$ is contained in $\mathfrak{u}_{n+1} = \text{span}(E_{ij}, j > i)$ for this basis. Conclude.
- (4) Adapt the previous argument to show the result when $\mathfrak{g}$ is the Lie algebra $\mathfrak{l}'_5$ from Exercise D. (First, why is it an extra case?) (Hint: check that $\mathfrak{l}'_5$ admits a grading in the positive integers).

Proposition and Definition 3.2. *Let $\mathfrak{g}$ be a nilpotent Lie algebra. Choose $\mathfrak{u}$ a Lie subalgebra of $\mathfrak{u}_n$ isomorphic to $\mathfrak{g}$. Then the set*

$$G = \exp(\mathfrak{u}) = \{\exp(X) : X \in \mathfrak{u}\}$$

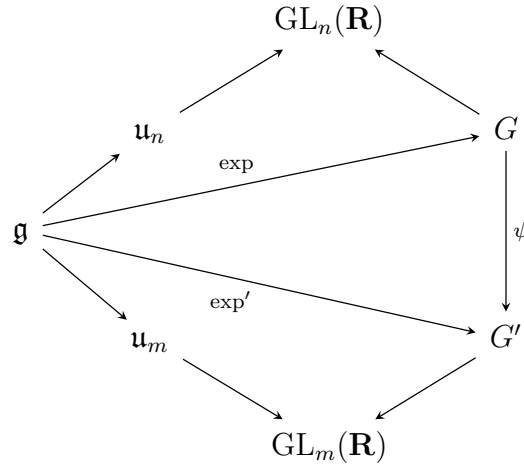
*is a simply connected, closed subgroup of $\text{GL}_n(\mathbf{R})$. $\exp$ is a homeomorphism onto its image. We define G to be the (simply connected) **Lie group** associated to $\mathfrak{g}$. This does not depend on $\mathfrak{u}$ in the following way: if $\mathfrak{u} \subseteq \mathfrak{u}_n$ and $\mathfrak{u}' \subseteq \mathfrak{u}_m$ as above, and $G = \exp(\mathfrak{u})$ and $G' = \exp(\mathfrak{u}')$, then there is a group isomorphism $\psi: G \rightarrow G'$ which is the restriction of a polynomial map $\text{GL}_n(\mathbf{R}) \rightarrow \text{GL}_m(\mathbf{R})$ with polynomial inverse, and commutes with the exponential maps. We also say that $\mathfrak{g}$ is the **Lie algebra of G** .*

From now on, we adopt the following convention: the same letter denotes a simply connected nilpotent Lie group and its Lie algebra, arabic uppercase and fraktur lowercase respectively. We declare that two simply connected nilpotent Lie groups G and H are **isomorphic as² Lie groups** if and only if $\mathfrak{g}$ and $\mathfrak{h}$ are isomorphic Lie algebras.

Remark 3.3 ()*. We will not prove Proposition-Definition 3.2 here; in a more systematic exposition of the theory, this should actually come after the BCH formula that we will soon state. Simply connected nilpotent Lie groups are real points of unipotent algebraic groups (See e.g. [Mil17, Chapter 14]) and there is a well-defined notion of polynomial map (with real coefficients) on a simply connected nilpotent Lie group which does not depend on the embedding in a linear group.

As a consequence of Proposition-Definition 3.2, $\exp : \mathfrak{g} \rightarrow G$ is well-defined independently of $\mathfrak{u}$ (See diagram on the right: ψ is the polynomial isomorphism with a polynomial inverse). And thus we will mostly ignore $\mathfrak{u}$ in the sequel. We denote the inverse of $\exp$ by $\log : G \rightarrow \mathfrak{g}$ and call it the **logarithm**. If G is embedded in $\exp(\mathfrak{u}_n)$ this is the restriction of the usual logarithm

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^n}{n-1} x^{n-1}.$$



In order to make sense of the statement of the next theorem, it is important to note that a formal series of Lie monomials over a finite alphabet has only finitely many terms of a given length, and so there is no convergence issue when substituting its indeterminates with elements of a nilpotent Lie algebra.

Theorem 3.4 (Baker-Campbell-Hausdorff series). *There exists a formal series of simple Lie monomials over the 2-letter alphabet, denoted H below, with coefficients in $\mathbb{Q}$, such that for any Lie subalgebra of $\mathfrak{u}_n$, it holds*

$$\log(\exp X \exp Y) = H(X, Y).$$

Moreover, the series $H(X, Y)$ cut to the terms of length at most four is

$$(3.1) \quad X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, X, Y] + [Y, Y, X]) - \frac{1}{24}[Y, X, X, Y].$$

See [God04, §6.6] for a proof which does not let the calculation of (3.1) as an exercise to the reader. There is a general version of the formula in non-nilpotent Lie algebras which deals with the convergence issues, see e.g. Bourbaki [Bou82, Chapter 2].

A very important consequence of the Baker-Campbell-Hausdorff series is that it allows us to write the multiplication law of a nilpotent group in **exponential coordinates** on its Lie algebra itself. Let us work out the important case of the Heisenberg Lie algebra $\mathfrak{heis} = \mathfrak{u}_3 = \mathfrak{l}_3$. Recall that it is generated with E_{12} , E_{23} and E_{13} and

²One can show that if the underlying topological groups of G and H are isomorphic as topological groups, then G and H are isomorphic as Lie groups. Beware that under Zorn's lemma, the abstract groups $\mathbb{R}$ and $\mathbb{R}^2$ are isomorphic and even isomorphic as $\mathbb{Q}$ -vector spaces.

that $[E_{12}, E_{23}] = E_{13}$, the brackets between other pairs of basis elements being zero. Writing (p, q, r) for the coordinates in this basis, we have

$$\log(\exp(p_1, q_1, r_1) \exp(p_2, q_2, r_2)) = \left(p_1 + p_2, q_1 + q_2, r_1 + r_2 + \frac{p_1 q_2 - p_2 q_1}{2} \right).$$

Exercise G. (1) Express the group law in terms of matrix coefficients on the Heisenberg group $\mathbf{Heis}(\mathbf{R}) = \exp(\mathfrak{u}_3) < \mathrm{GL}_3(\mathbf{R})$, and recover the expression of the group law in exponential coordinates. (Hint : remember about the series expansion of $\log$.)

- (2) Sketching the same method with another group, explain why the coefficient $\pm \frac{1}{12}$ appears in front of the Lie monomials of length 3 in the BCH series. (You may want to use a computer algebra system to help with the matrix computations.)

From now on, given a nilpotent Lie algebra $\mathfrak{g}$ and $x, y \in \mathfrak{g}$, we will denote

$$x * y := \log(\exp(x) \exp(y)).$$

We use lowercase for x and y to emphasize that we think of them as group elements, even if they are formally elements of the Lie algebra.

Proposition 3.5. *Let $\mathfrak{g}$ be a nilpotent Lie algebra. Equip it with a Lebesgue measure λ that is invariant under vector space translation. Then λ is also a Haar measure on the nilpotent group $(\mathfrak{g}, *)$.*

Proof. We need to show that for every $x \in \mathfrak{g}$, the left multiplication $L_x : y \mapsto x * y$ has its Jacobian determinant identically equal to 1. Equip $\mathfrak{g}$ with an adapted basis $(e_1, \dots, e_n)$. Let $x = (x_1, \dots, x_s)$ and $y = (y_1, \dots, y_s)$ in the corresponding coordinates. It follows from the BCH formula that

$$x * y = (x_1 + y_1, x_2 + y_2, x_3 + y_3 + f_3(x_1, x_2, y_1, y_2), \dots, x_n + y_n + f_n(x_{\hat{n}}, y_{\hat{n}}))$$

where f_i are polynomials for $i = 3, \dots, n$ and $x_{\hat{n}}$ denotes $(x_1, \dots, x_{n-1})$. Thus

$$\frac{\partial(L_x)_j}{\partial y_i}(y) = \begin{cases} 0 & i > j \\ 1 & i = j \\ \frac{\partial f_j}{\partial y_i}(x_1, \dots, x_{j-1}, y_1, \dots, y_{j-1}) & i < j. \end{cases}$$

The Jacobian matrix of the left multiplication by x is triangular with 1s on the diagonal, hence its determinant is 1. $\square$

Definition 3.6. Given a simply connected nilpotent Lie group G , we denote by G_∞ the nilpotent Lie group with Lie algebra $\mathfrak{g}_\infty$. We call it the associated Carnot group of G . If G and G_∞ are isomorphic as Lie groups (which occurs if and only if $\mathfrak{g}$ and $\mathfrak{g}_\infty$ are isomorphic Lie algebras), we say that G is a Carnot group.

Proposition 3.7. *Let $G = (\mathfrak{g}, *)$ be a Carnot group, and let $\{t \bullet -\}$ be the 1-parameter group of dilations of $\mathfrak{g}$. For every $t \in \mathbf{R}_{>0}$, $x \mapsto t \bullet x$ defines a Lie group automorphism of G . We call it the Carnot dilation of G .*

Proof. Using the BCH series,

$$t \bullet (x * y) = t \bullet H(x, y) = H(t \bullet x, t \bullet y) = (t \bullet x) * (t \bullet y).$$

This proves that $t \bullet -$ is a Lie group endomorphism. Its inverse is $t^{-1} \bullet -$. $\square$

Definition 3.8. A Lie group homomorphism $\Psi: G \rightarrow H$ between Carnot groups G and H is a **Carnot group homomorphism** if $\Psi(t \bullet x) = t \bullet (\Psi(x))$ for all $x \in G$ and $t \in \mathbf{R}_{>0}$.

It is worth noting, again, that in $\mathbf{R}^d$, the fact that automorphisms commute with the dilation is a consequence of their mere linearity, but in general, it's not.

Remark 3.9 ()*. Although this course is not primarily concerned with finitely generated groups, let us mention at this stage that a theorem of Mal'cev ensures that, as soon as the nilpotent Lie algebra $\mathfrak{g}$ admits a basis $(e_1, \dots, e_n)$ with

$$[e_i, e_j] = \sum_k c_{i,j}^k e_k$$

where all the coefficients are rational, G admits a co-compact lattice, which is a finitely generated torsion-free nilpotent group [Mal51]. Explicit presentations of these lattices can be worked out from the BCH series. For instance, the Heisenberg group has a lattice whose presentation is

$$\langle x, y, z \mid [x, z] = [y, z] = 1, [x, y] = z \rangle.$$

This theorem of Mal'cev has a converse which is an ancestor of Theorem 0.1 and which ensures that every finitely generated torsion-free nilpotent group is a lattice in a simply connected Lie group (We refer to Theorem A.6 for a precise statement). This lattice is realized as $\mathbf{Heis}(\mathbf{Z})$. More generally, lattices in nilpotent Lie groups are virtually the integer points of unipotent algebraic groups defined over $\mathbf{Q}$; different $\mathbf{Q}$ -groups with the same real points will yield non-commensurable lattices in a same simply connected nilpotent Lie group.

4. HOMOGENEOUS NORM AND GROWTH

Let $\mathfrak{g}$ be a nilpotent Lie algebra. Let $\mathfrak{g}_\infty$ be its associated Carnot graded Lie algebra. Let V be the vector space underlying $\mathfrak{g}_\infty$. Write $V_i = (\mathfrak{g}_\infty)_i$.

Let $\mathfrak{L}: \mathfrak{g} \rightarrow \mathfrak{g}_\infty$ be a linear isomorphism respecting the lower central series filtrations (any choice of adapted basis on $\mathfrak{g}$ gives rise to such an isomorphism, by mapping every basis element to its image modulo the first term in the central series where it isn't). On the underlying vector space V of $\mathfrak{g}_\infty$ we have :

- (1) The Carnot bracket $\{X, Y\}_\infty$
- (2) The original bracket $[X, Y] = \mathfrak{L}[\mathfrak{L}^{-1}X, \mathfrak{L}^{-1}Y]$
- (3) The group law $(x, y) \mapsto x *_\infty y$ coming from the BCH series evaluated on the graded bracket:

$$x *_\infty y = x + y + \frac{1}{2}\{x, y\} + \frac{1}{12}(\{x, x, y\} + \{y, y, x\}) + \dots$$

- (4) The group law $(x, y) \mapsto x * y$ coming from the BCH series evaluated on the original bracket:

$$x * y = x + y + \frac{1}{2}[x, y] + \frac{1}{12}([x, x, y] + [y, y, x]) + \dots$$

Note that $-x$ is unambiguous: it is equal to x^{-1} for $*_\infty$ and $*$. (Beware however that in general $-(x * y) \neq (-x) * y$ and also $-(x *_\infty y)$.)

Definition 4.1. Fix a norm on V . Let $x \in \mathfrak{g}_\infty$; let $x = \underline{x}_1 + \cdots + \underline{x}_s$ where $\underline{x}_i \in V_i$ (We underline the “components” $\underline{x}_i$ to emphasize that these are still vectors). Define the **homogeneous norm** of x as

$$\lfloor x \rfloor := \sup_i \|\underline{x}_i\|^{1/i}.$$

Theorem 4.2 (Guivarc’h 1973, [Gui73]). *Let d be a proper, $*$ -invariant geodesic distance on $\mathfrak{g}_\infty$. Then there exists $\lambda \geq 1$ and $c \geq 0$ such that $\frac{1}{\lambda} \lfloor x \rfloor - c \leq d(0, x) \leq \lambda \lfloor x \rfloor + c$ for all $x \in \mathfrak{g}$.*

Remark 4.3. The norm that Guivarc’h actually defines is $\lfloor x \rfloor = \|x_1\| + \cdots + \|x_s\|^{1/s}$.

Note that the theorem also applies to any $*$ -invariant distance. A variant of this is known as the “ball-box theorem” [LD25] because it allows to put balls in between boxes, as we shall see. Let’s first give an application.

Corollary 4.4 (Bass-Guivarc’h formula, modern version). *Let G be a CGLC group with polynomial growth and let S be a compact generating set of G . Then there are constants C_+ and C_- such that for n large enough*

$$C_- n^d \leq \text{vol}(S^n) \leq C_+ n^d.$$

Moreover,

$$(4.1) \quad d = \sum_{i=1}^s i \dim C^i \mathfrak{g} / C^{i+1} \mathfrak{g} = \sum_{i=1}^{\infty} \dim C^i \mathfrak{g}$$

where $\mathfrak{g}$ is the Lie algebra of the connected nilshadow of G .

Proof. Define a box $\mathbf{B}(r) = \{x \in V : \lfloor x \rfloor \leq r\}$ where the homogeneous norm $\lfloor - \rfloor$ comes from a ℓ^∞ norm in an adapted basis. The Haar/Lebesgue volume of a box is

$$\text{vol}(\mathbf{B}(r)) = \prod_{i=1}^s \text{vol}_{\dim V_i} B_{\ell^\infty}(0, r^i) = 2^{\dim V} \prod_{i=1}^s r^{i \dim V_i}.$$

By Theorem 4.2, $\text{vol}(S^n)$ is of the order of $\text{vol}(\mathbf{B}(n))$. □

Exercise H. In the two latter questions of this exercise, you are allowed to use either Losert’s theorem from the first lecture, or Gromov’s polynomial growth theorem together with Mal’cev Theorem A.6 from the appendix.

- (1) Compute the order of polynomial growth of the Lie groups having Lie algebras $\mathfrak{heis}$, $\mathfrak{l}_4$, and $\mathfrak{g}_{4,3}$.
- (2) Let $\Gamma = \langle S \rangle$ be a finitely generated group. Assume that $|S^n| = O(n^\pi)$. Show that Γ has a free abelian finite-index subgroup.
- (3) Let Γ be a finitely generated group quasiisometric to $\mathbf{Z}^d$ for $d \in \{1, 2, 3\}$. Show that Γ has a finite-index subgroup isomorphic to $\mathbf{Z}^d$.

Example 4.5. Let us compute the order of growth of L_{s-1} (which appeared briefly on the board towards the end of Matt’s first talk on Monday). We have

$$(4.2) \quad d = 2 + 2 + 3 + \cdots + s - 1 = 1 + \frac{s(s-1)}{2}.$$

This is the maximal order of growth of a $(s+1)$ -dimensional connected nilpotent Lie group.

Remark 4.6. Let G be a simply connected nilpotent Lie group. Then G and G_∞ have the same order of polynomial growth.

Theorem 4.7 (Goodman, Cornulier [Goo77, Cor19]). *Let $\mathfrak{g}$ be a nilpotent Lie algebra. There exists $e = e(\mathfrak{g}) \in [0, 1)$ such that $\lfloor x * y - x *_\infty y \rfloor = O(1 + \lfloor x \rfloor^e + \lfloor y \rfloor^e)$.*

Cornulier's version includes an upper bound on $e(\mathfrak{g})$ which is in general better than the bound $e(\mathfrak{g}) \leq 1 - 1/s$ coming out of Goodman's proof. We will only prove the theorem when $s = 3$ which is the first case of interest (when $s = 2$, G is Carnot and the statement follows with $e = 0$). The case $s = 4$ is the first where Cornulier's improvement on Goodman's become noticeable.

Proof when $s = 3$. Assume $s = 3$. For a vector $X \in V$, denote X_i its projection to V_i in the direct sum decomposition $V = V_1 \oplus \dots \oplus V_s$. Start noticing that for $X, Y \in V_1$ we have $[X, Y] \in C^2\mathfrak{g} = V_2 \oplus V_3$ but $\{X, Y\} \in V_2$. This is, in fact, the only difference between the two brackets. Precisely we compute that

$$[X, Y] - \{X, Y\} = [X, Y]_3$$

Then

$$\begin{aligned} \lfloor x * y - x *_\infty y \rfloor &= \left\lfloor \frac{1}{2} [x_1, y_1]_3 \right\rfloor \lesssim (\|x_1\| \|y_1\|)^{1/3} \lesssim (\max(\|x_1\|, \|y_1\|))^{2/3} \\ &\leq \lfloor x \rfloor^{2/3} + \lfloor y \rfloor^{2/3}. \end{aligned}$$

This is Goodman's estimate with $e = 2/3 = 1 - 1/s$. $\square$

5. SUBLINEAR BILIPSCHITZ EQUIVALENCE

Definition 5.1. Let X and Y be metric spaces; fix basepoints in X and Y and denote distances in X and Y by d and distances to basepoint by $|\cdot|$. Let $\gamma \in [0, 1)$. Let $f: X \rightarrow Y$. We say that f is a $O(r^\gamma)$ -sublinear bilipschitz equivalence if for some (any) left-invariant proper roughly geodesic distances on G and H ,

$$\frac{1}{\lambda} d(x, y) - O(1 + |x|^\gamma + |y|^\gamma) \leq d(f(x), f(y)) \leq \lambda d(x, y) + O(1 + |x|^\gamma + |y|^\gamma)$$

and $d(y, f(X)) = O(1 + |y|^\gamma)$.

Remark 5.2. For $e = 0$, this is exactly the definition of a quasiisometry.

Sublinear bilipschitz equivalences were introduced by Cornulier, at first without a name [Cor08, Cor11, Cor19], in order to describe the asymptotic cones of Lie groups.

Remark 5.3 ()*. Sublinear bilipschitz equivalences are *not* coarse equivalences. However, there exist sublinear coarse structures that they preserve [Pal22]. In that sense, they are “sublinear coarse equivalences”.

Exercise I. Let X, Y, Z be pointed metric spaces. Let $f: X \rightarrow Y$ be a $O(r^\gamma)$ -SBE and let $g: Y \rightarrow Z$ be a $O(r^{\gamma'})$ -SBE. Show that $g \circ f: X \rightarrow Z$ is a $O(r^{\max(\gamma, \gamma')})$ -SBE.

Theorem 5.4 (Cornulier). *Let G and H be simply connected nilpotent Lie groups. If $G_\infty \simeq H_\infty$, then there exists $e \in [0, 1)$ (depending on the pair (G, H)) such that G and H are $O(r^e)$ -SBE*

Proof. Equip the vector space V underlying $\mathfrak{g}_\infty$ with the group law $*_\infty$, and, via the data of supplementary subspaces to the lower central series, with the group law $*$ of G . Let d_∞ and d denote proper geodesic distances on V which are respectively $*_\infty$ and $*$ -invariant. Using successively Guivarc'h's, Goodman's, and then again Guivarc'h's estimates we get

$$\begin{aligned}
d(x, y) &= d(0, -x * y) && \text{since } d \text{ is } * \text{-invariant} \\
&\leq \lambda[(-x) * y] + c && \text{by Theorem 4.2} \\
&\leq \lambda[(-x) *_\infty y] + \lambda[(-x) * y - (-x) *_\infty y] + c && \text{by triangle inequality} \\
&\leq \lambda[(-x) *_\infty y] + \lambda(|x|^e + |y|^e) + c && \text{by Theorem 4.7} \\
&\leq \lambda^2 d_\infty(0, (-x) *_\infty y) + \lambda(|x|^e + |y|^e) + \lambda c + c && \text{by Theorem 4.2} \\
&= \lambda^2 d_\infty(x, y) + O(|x|^e + |y|^e). && d_\infty \text{ is } *_\infty \text{-invariant}
\end{aligned}$$

In the previous series of inequalities, we did not specify whether $|x|$ denotes $d(1, x)$ or $d_\infty(1, x)$. It turns out that it does not matter thanks to Guivarc'h estimates. A similar argument shows that $d_\infty(x, y) \leq \lambda^2 d(x, y) + O(|x|^e + |y|^e)$. This proves that G and G_∞ are $O(r^e)$ -sublinear bilipschitz equivalent. Similarly, H and H_∞ are $O(r^{e'})$ -sublinear bilipschitz equivalent. $\square$

Using Cornulier's theorem we can recover (a weak form of) the main result of Pansu's 1982's "first thesis" [Pan83]³.

Definition 5.5. Say that a sequence of compact metric spaces (X_n) **converges in Gromov-Hausdorff's sense** to a metric space X_∞ if

$$\inf_{\iota, \iota'} d_H(\iota(X_n), \iota'(X_\infty)) \rightarrow 0$$

where the infimum is taken over all isometric embeddings ι, ι' of X_n and X_∞ in compact metric spaces Z .

Proposition and Definition 5.6 (Consequence of Gromov's compactness theorem). *Let G be a simply connected nilpotent Lie group, equipped with a left-invariant Riemannian metric d . There exists a proper metric space (X_∞, Δ) such that for all $r > 0$, the sequence of metric spaces*

$$X_n = \left(B_d(1_G, rn), \frac{d}{n} \right).$$

*has a subsequence that converges to $B_\Delta(X_\infty, r)$ in Gromov-Hausdorff's sense. We call X_∞ a (pointed) **asymptotic cone** of G .*

By proper metric space we mean that the closed balls are compact. The proof (that we skip) is not very hard, but it is instructive; among the main ingredients of Gromov's proof, this is arguably the easiest to convey. The conclusion holds as well for complete Riemannian manifolds under a condition of positivity of Ricci curvature (both condition implying a *doubling* property). See [Gro99]. Using a diagonal extraction we can assume that the same subsequence converges in Gromov-Hausdorff sense for all r .

Remark 5.7 ().* Those who know the subject may be aware of the following facts that are not present in our exposition.

³Pansu's first thesis is the "thèse de troisième cycle" which corresponds to today's doctorate. The second thesis (defendend in 1987) that we evoke after is the "thèse d'état" which corresponds to today's habilitation.

- (1) There is a way to define asymptotic cones avoiding the extraction of subsequences, using a non-principal ultrafilter. This was realized by van den Dries and Wilkie [vdDW84] and then adopted by Gromov and geometric group theorists [Gro93], cf. Anthony's second talk on Tuesday, and Oussama's talk on Thursday.
- (2) It follows from the actual version of Pansu's theorem that asymptotic cones of nilpotent Lie groups are *unique* and that there is no need of passing to a subsequence. We don't want to use this because our current goal is to prove a weak version of Pansu's theorem. We will come back to this a bit later.

Lemma 5.8. *Let G and H be simply connected nilpotent groups. If there exists a $O(r^e)$ -sublinear bilipschitz equivalence $f: G \rightarrow H$ for some $e \in [0, 1)$, then for any pointed asymptotic cone X_∞ of G there is a pointed asymptotic cone Y_∞ of H and a bilipschitz homeomorphism $X_\infty \rightarrow Y_\infty$.*

Proof. Set X_n as previously, namely $X_n = (B_d(1_G, rn), d/n)$. Consider

$$f_n: X_n \longrightarrow \left(H, \frac{d_H}{n}\right).$$

and let $x_1, x_2 \in X_n$. Then

$$\frac{d_H}{n}(f(x_1), f(x_2)) \leq \lambda \frac{d_G}{n}(x_1, x_2) + 2 \frac{(rn)^e}{n}.$$

Using the lower estimate on $d(f(x_1), f(x_2))$, we show that X_n and $f_n(X_n)$ are (λ, cn^{1-e}) -quasiisometric for some $c > 0$. An Ascoli-Arzelà type argument finishes the proof. $\square$

Note that the lemma holds with SBE replaced by quasiisometry. If f is a quasiisometry, it induces bilipschitz homeomorphisms between pairs of asymptotic cones with *moving observation centers*. We first state the weak version of Pansu's theorem and then explain the new words inside it.

Theorem 5.9 (Weak version of “Pansu's first thesis”). *Let G be a simply connected nilpotent Lie group. Any asymptotic cone of G is bilipschitz homeomorphic to G_∞ endowed with a **Carnot-Carathéodory** distance.*

We now explain what a Carnot-Carathéodory distance is. Let $\mathfrak{g}$ be a nilpotent Lie algebra. Fix a norm $\| - \|$ on $\mathfrak{g}_1 = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ and equip the Carnot group G_∞ with the following distance (where CC stands for Carnot-Carathéodory): for every $x, y \in \mathfrak{g}_\infty$,

$$(5.1) \quad d_{\text{CC}}(x, y) = \inf \int \|L_{\gamma(t)}^* \gamma'(t)\| dt,$$

the infimum being taken on piecewise C^1 -paths⁴ from x to y that such that $\gamma'(t) \in (L_{\gamma(t)})_* \mathfrak{g}_1$ for all t (See figure 2). One can show that d_{CC} is a left-invariant, proper geodesic metric on G (see [LD25]). Thus d_{CC} can play the role of a metric d_∞ appearing in the proof of Theorem 5.4.

Lemma 5.10. *Let G be a Carnot group equipped with a Carnot-Carathéodory metric d_{CC} . For all $t \in \mathbf{R}_{>0}$ and $x, y \in G$,*

$$(5.2) \quad d_{\text{CC}}(t \bullet x, t \bullet y) = t d_{\text{CC}}(x, y)$$

⁴It is not very natural to take piecewise C^1 -paths (and this certainly does not ease the proof of the claims we make here). However, it does not make a difference in the end (for non-trivial reasons, see [LD11, Theorem 1.2]) and it is simpler for the definition.

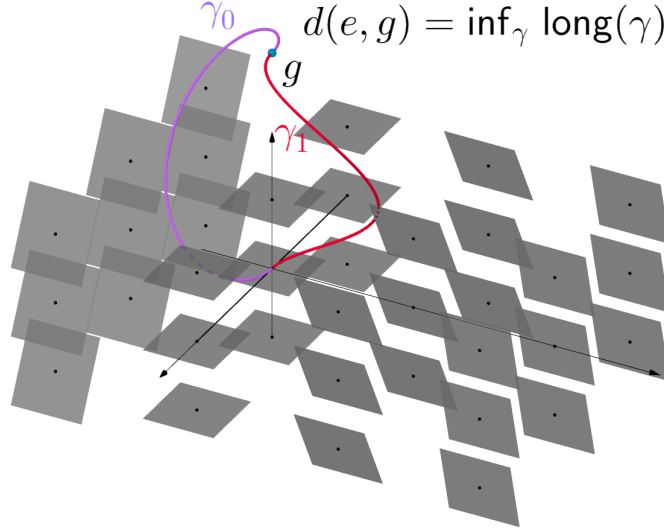


FIGURE 2. The Carnot-Carathéodory distance. The rectangles represent the horizontal distribution.

Proof. The proof is not hard once we give a name to the relevant object. This object is the subbundle $\mathcal{H}$ whose fiber above x is $(L_{\gamma}(t))_{*}\mathfrak{g}_1$. The dilation $t \bullet -$ naturally extends to this bundle and sends $\mathcal{H}_x$ into $\mathcal{H}_{t \bullet x}$ multiplying norms by t . Especially, whenever γ is a curve between x and y that is tangent to $\mathcal{H}$, $t \bullet \gamma$ is a curve between $t \bullet x$ and $t \bullet y$ and it has length

$$\int \|L_{t \bullet \gamma(t)}^{*}(t \bullet \gamma)'(t)\| dt = t \int \|L_{\gamma(t)}^{*}\gamma'(t)\| dt.$$

This finishes the proof. $\square$

Lemma 5.11. *Let G be a Carnot group and let d_{CC} be a Carnot-Carathéodory metric on it. Any asymptotic cone of (G, d_{CC}) is isometric to (G, d_{CC}) .*

Proof. By the previous lemma, $x \mapsto t \bullet x$ is a dilation of the Carnot-Carathéodory distance d_{CC} . It follows that all the spaces

$$X_n = \left(B_{d_{CC}}(1_G, rn), \frac{d_{CC}}{n} \right)$$

are all isometric, and at zero Gromov-Hausdorff distance. It follows that they are equal to any Gromov-Hausdorff limit of a subsequence. $\square$

We can now assemble all the ingredients and prove the theorem.

Proof of Theorem 5.9. By Theorem 5.4, G and G_{∞} are $O(r^e)$ -sublinear bilipschitz equivalent. Let X_{∞} be an asymptotic cone of G . By Lemma 5.8, there is some asymptotic cone of G_{∞} , say Y_{∞} to which X_{∞} is bilipschitz homeomorphic. Since d_{CC} is a proper geodesic metric on G_{∞} which is its own asymptotic cone, X_{∞} is bilipschitz homeomorphic to d_{CC} . $\square$

Theorem 5.12 (One part of Pansu's second thesis from the SBE viewpoint). *If there exists $e \in [0, 1)$ such that G and H are $O(r^e)$ -SBE, then $G_{\infty} \simeq H_{\infty}$.*

A few words on the proof. (*) First, note that we may assume without loss of generality that G and H are Carnot, replacing them with their associated Carnot groups if needed.

The heart of the argument is based on the observation that the dilation of a Carnot group is in many ways similar to the scalar dilation of a euclidean space. We have seen that this distance has the following fundamental property.

$$(5.3) \quad d_{CC}(t \bullet x, t \bullet y) = t d_{CC}(x, y)$$

for every $t \in \mathbf{R}_{>0}$. Say that a map $\phi: G \rightarrow H$ is P -differentiable at $p \in G$ if

$$\forall y \in G, D_P \phi_x(y) := \lim_{t \rightarrow 0} \frac{1}{t} \bullet (\phi(x * t \bullet y) \phi(x)^{-1})$$

exists for all $y \in G$ and that the convergence to the limit above is uniform on every compact set of G . If G and H are abelian and ϕ is continuously differentiable, then $D_P \phi_x$ is just the usual differential of ϕ at x . Moreover, the differential $D_P \phi_x$ is a Lie group homomorphism intertwining the Carnot dilations, very much like the usual differential of a differentiable map at a point is a linear map on the tangent space.

Theorem 5.13 (Pansu’s Rademacher theorem, [Pan89]). *Any lipschitz map between d_{CC} distances on Carnot groups is almost everywhere (with respect to a Haar measure) P -differentiable.*

The proof of Theorem 5.12 uses this in the following way. The differential of $\text{Cone}(f): G \rightarrow H$ is almost everywhere a (Carnot) Lie group isomorphism. This implies that the groups G and H are isomorphic. $\square$

Let us finish our discussion of asymptotic cones of nilpotent Lie groups by coming back on the uniqueness issue. Here is a stronger form of Theorem 5.4.

Theorem 5.14 (Pansu). *Let G be a simply connected nilpotent Lie group. The asymptotic cone of any proper left-invariant geodesic distance d on G is G_∞ endowed with a Carnot-Carathéodory distance.*

This theorem states that the asymptotic cone is unique. (The detailed version also says which norm one should put on $\mathfrak{g}_1$.) We want to point out that the uniqueness is an essential difference with Theorem 5.4. In fact this uniqueness, together with the two results below and Pansu’s differentiation theorem, are sufficient to deduce the slightly stronger statement.

Fact 5.15 (Gromov⁵ [Gro81]). *The asymptotic cone of a simply connected nilpotent Lie group is a locally compact group, equipped with a geodesic metric.*

Fact 5.16 (Le Donne [LD15, Theorem 1.1]). *Carnot groups with d_{CC} metrics are the only metric spaces that are locally compact, geodesic, isometrically homogeneous, and admit a 1-parameter subgroup of metric dilations.*

In the statement of Fact 5.16, “isometrically homogeneous” means that there is a transitive subgroup of isometries. We warn the reader that although Le Donne’s proof of Fact 5.16 is quite short, it uses Montgomery-Zippin-Gleason’s solution to Hilbert’s fifth problem.

⁵This is one step in the proof of the polynomial growth theorem and it is originally concerned with finitely generated groups, but there is no significative difference.

Proof of Theorem 5.14 admitting the uniqueness of asymptotic cone of (G, d) . We already know that *some* asymptotic cone of (G, d) is bilipschitz homeomorphic to (G_∞, d_{CC}) . Let X_∞ be that asymptotic cone. By uniqueness of X_∞ , it has a one-parameter of dilations. By Fact 5.15 and Le Donne's characterisation of Carnot groups, X_∞ is some (H_∞, d_{CC}) . By Theorem 5.12 G_∞ and H_∞ are isomorphic as Lie groups. Hence **the** asymptotic cone of (G, d) is (G_∞, d_{CC}) for some d_{CC} . $\square$

We end this section with an observation by Cornulier that having polynomially bounded growth is an SBE invariant. Indeed, it can be characterized at the level of asymptotic cones in the following way.

Theorem 5.17 (See e.g. [Cor19, Proposition 3.2 and Corollary 3.5]). *Let G be a CGLC group. G has polynomial growth if and only if all its asymptotic cones are proper.*

Note that a metric space bilipschitz homeomorphic to a proper metric space is proper as well.

Exercise J. (*)

- (1) Let Γ be a finitely generated group $O(r^e)$ -SBE to $\mathbf{Z}^d$ for some $e \in [0, 1)$ and $d \in \mathbf{N}$. Show that Γ has a finite-index subgroup isomorphic to $\mathbf{Z}^d$.
- (2) Let Γ be a finitely generated group QI to $\Lambda = \mathbf{Heis}(\mathbf{Z}[\sqrt{2}])$ (first, make sense of this statement). Show that Γ has a finite-index subgroup isomorphic to a lattice in $L = \mathbf{Heis}(\mathbf{R})^2$. Do Γ and Λ have isomorphic finite-index subgroups?

6. A QUANTITATIVE PERSPECTIVE

The open problem from the beginning is not solved. But along the way, we have seen that some non-isomorphic nilpotent Lie groups are SBE. This leaves some room for the following.

Imprecise question: Given two simply connected, non-isomorphic nilpotent groups, to what extent are they not quasiisometric?

This is a softer question on which we can hope to make some gradual progress. Pansu and Cornulier's theorems suggest to make the imprecise question precise in the following way.

Question 6.1. Let N and N' be simply connected nilpotent groups with $N_\infty \simeq N'_\infty$ as Lie groups. What is the range of $e \in [0, 1)$ such that N and N' are $O(r^e)$ -SBE?

Definition 6.2. Let $\mathfrak{g}$ and $\mathfrak{h}$ be nilpotent Lie algebra, and let $\phi : Z(\mathfrak{g}) \rightarrow Z(\mathfrak{h})$ be a Lie group homomorphism. We denote $\mathfrak{g} \times_\phi \mathfrak{h}$ and call central product of $\mathfrak{g}$ and $\mathfrak{h}$, the Lie algebra

$$\mathfrak{g} \times_\phi \mathfrak{h} := (\mathfrak{g} \times \mathfrak{h})/\mathfrak{w},$$

where $\mathfrak{w} = \{(X, \phi(X)) : X \in Z(\mathfrak{g})\}$.

Example 6.3. Recall from Example 2.3 that

$$\mathfrak{g}_{4,3} = \text{span}(X_1, X_2, X_3, Y_1, Y_2, Z).$$

We have $\mathfrak{g}_{4,3} = \mathfrak{l}_4 \times_\phi \mathfrak{l}_3$, where $\phi : Z(\mathfrak{l}_4) = \langle X_4 \rangle \rightarrow Z(\mathfrak{l}_3) = \langle X_3 \rangle$, $\phi(X_4) = -X_3$.

Proposition and Definition 6.4. *Let $\mathfrak{g}$ and $\mathfrak{h}$ be nilpotent Lie algebras with one-dimensional centers. Assume that $\mathfrak{g}$ or $\mathfrak{h}$ is Carnot. The isomorphism class of $\mathfrak{g} \times_{\phi} \mathfrak{h}$ does not depend on the choice of a linear isomorphism $\phi : Z(\mathfrak{g}) \rightarrow Z(\mathfrak{h})$. We denote it $\mathfrak{g} \times_Z \mathfrak{h}$.*

Proof. Without loss of generality, $\mathfrak{h}$ is Carnot; let s be its nilpotency class. Let ϕ_1 and ϕ_2 be isomorphisms from $Z(\mathfrak{g})$ to $Z(\mathfrak{h})$. Because these are 1-dimensional, there is $\alpha \in \mathbf{R} \setminus \{0\}$ such that $\phi_2 = \alpha \phi_1$. First assume that $\alpha > 0$, and let $t = \alpha^{1/s}$. Denote $\mathfrak{w}_i = \{(X, \phi_i(X)) : X \in Z(\mathfrak{g})\} \subseteq \mathfrak{g} \times \mathfrak{h}$ for $i = 1, 2$. Define

$$\begin{aligned} \Psi : \mathfrak{g} \times_{\phi_1} \mathfrak{h} &\rightarrow \mathfrak{g} \times_{\phi_2} \mathfrak{h}, \\ (X, Y) \bmod \mathfrak{w}_1 &\mapsto (X, t \bullet Y) \bmod \mathfrak{w}_2. \end{aligned}$$

This is a Lie algebra isomorphism (technically, the map $(X, Y) \mapsto (X, t \bullet Y)$, which is clearly an isomorphism, descends to a well-defined morphism between the quotients; the reader can draw the commuting diagram to convince themselves). If $\alpha < 0$ and s is odd, one can still define this isomorphism. If $\alpha < 0$ and s is even: note that $\mathfrak{h}$ admits a Lie algebra automorphism ε given by $X \mapsto (-1)^i X$ for $X \in \mathfrak{h}_i$. One then gets an isomorphism after sending $(X, Y) \bmod \mathfrak{w}_1$ to $(X, \frac{1}{t} \bullet \varepsilon(Y)) \bmod \mathfrak{w}_2$. $\square$

We denote by $G \times_Z H$ the simply connected nilpotent Lie group with Lie algebra $\mathfrak{g} \times_Z \mathfrak{h}$. It is s -nilpotent where s is the maximum of the nilpotency classes of G and H .

Given a simply connected nilpotent Lie group G denote by $\underline{e}(G)$ the infimal exponent e such that G and G_{∞} are $O(r^e)$ -SBE. An estimate on $\underline{e}$ is a contribution to Question 6.1.

Theorem 6.5 (Combination of [Cor19] and [GMLIP23]). *For every $s \geq 3$*

$$\frac{1}{2s} \leq \underline{e}(L_s \times_Z L_3) \leq \frac{2}{s-1}.$$

The upper inequality is proved in [Cor19, Section 6]. The lower inequality involves the computation of the **Dehn functions** of the groups $L_s \times_Z L_3$. A large part of the methods of [GMLIP23] were developed in [LIPT23]. The suggestion to evaluate the Dehn function of $\mathfrak{g}_{4,3}$ in relation to this problem was made in [Cor19].

Remark 6.6 ()*. There is another approach to the quasiisometric classification of nilpotent groups, initiated by Shalom [Sha04] and pursued in [Sau06, GK21, AY25]. See Theorem A.16 for a statement. We refer to Roman Sauer's 2023 course in Pisa on this topic: [Lectures 1 and 2](#) (starting at 3:35:46) and [Lecture 3](#) (starting at 3:39:36).

APPENDIX A. LATTICES IN NILPOTENT LIE GROUPS

In the motivational §0 we have seen that one can associate a simply connected nilpotent Lie group (the connected nilshadow) to every compactly generated locally compact group of polynomial growth. This applies in particular to finitely generated groups of polynomial growth, which are virtually nilpotent by Gromov's theorem. In this case, the construction of the connected nilshadow proceeds from a construction of Mal'cev that we evoked in the end of §3 and it is much older than Losert's theorem. The aim of this appendix is to provide some details on this construction of Mal'cev. We want to emphasize in particular that there are (many) simply connected nilpotent Lie groups without lattices, and that the QI classification problem is indeed more vast

when posed for nilpotent Lie groups rather than when it is posed for finitely generated groups of polynomial growth (we will recall a proof of this based on a Baire category argument).

In this appendix, the Lie algebras can have a ground field different from $\mathbf{R}$ (but always of characteristic 0). We assume more knowledge on Lie groups and Lie algebra than in the rest of the text.

Zariski density of lattices. We refer to Chapter II of Raghunathan's book [Rag72] for the proofs of the following results. In nilpotent Lie groups, all lattices are uniform⁶.

Definition A.1. Let G be a nilpotent Lie group and $E \subset G$ a closed subset. We say that E is **Zariski-dense** in G if there exists a faithful unipotent representation $\rho : G \rightarrow \mathrm{GL}(n, \mathbf{R})$ such that $\rho(E)$ and $\rho(G)$ have the same Zariski closure in $\mathrm{GL}(n, \mathbf{C})$.

A nilpotent Lie group G admits faithful unipotent representations as soon as it is simply connected. In this case, it is shown that it is equivalent, in the previous definition, to require that $\rho(E)$ and $\rho(G)$ have the same Zariski closure for any such representation ρ . Moreover,

Theorem A.2 ([Rag72, th 2.1]). *For a closed subgroup Γ of a simply connected nilpotent Lie group G , the following are equivalent:*

- (i) Γ is Zariski-dense.
- (ii) G/Γ is compact.
- (iii) G/Γ admits a finite left-invariant measure.
- (iv) G is minimal among its closed connected subgroups containing Γ .

This theorem has several important corollaries. For example, [Rag72, cor 2.3], if Γ is a discrete subgroup of G , it is finitely generated, its rank is well defined, and the dimension of its Zariski closure $\tilde{\Gamma}$ in G is equal to it [Rag72, th 2.10]. In particular, if Γ is a lattice, its rank is the dimension of the ambient group.

Finally, homomorphisms from the group are entirely determined by their restriction to a lattice, and there is no restriction on this:

Proposition A.3 (strong rigidity). *Let G and G' be simply connected nilpotent Lie groups, and let $\Gamma < G$ be a lattice. Every continuous homomorphism $\Gamma \rightarrow G'$ extends to a unique global homomorphism. In particular, every automorphism of Γ extends to a unique automorphism of G .*

Remark A.4 (*). The uniqueness in the previous proposition is related to the Zariski-dense character of Γ . Let us detail an example to indicate how different the situation is when G is $\mathrm{SL}(2, \mathbf{R})$. Let the group $G = \mathrm{SL}(2, \mathbf{R})$. Let Σ be a hyperbolic surface of genus $g \geq 2$, and $\Gamma = \pi_1 \Sigma$. The space of faithful representations of Γ in G modulo conjugation is that of hyperbolic structures on Σ ; it is of dimension $6g - 6$ according to Fenchel-Nielsen theory, hence much larger than $\mathrm{Aut}(\mathrm{SL}(2, \mathbf{R}))$. On the other hand, the group $\mathrm{SL}(2, \mathbf{Z})$ is Zariski-dense in G , although it is not even cocompact, unlike Γ .

Corollary A.5 (Translation in terms of commensurability). *Let Γ_1 and Γ_2 be two lattices in a simply connected nilpotent Lie group G . The following are equivalent:*

- (i) Γ_1 and Γ_2 admit finite index subgroups that are isomorphic: Γ_1 and Γ_2 are commensurable, [dlH00, IV.27].

⁶This holds more generally in all locally compact groups that are amenable, [Gui73, lemma 1.10].

- (ii) *There exists $g \in G$ such that $g\Gamma_1 g^{-1} \cap \Gamma_2$ is of finite index in both Γ_1 and Γ_2 : Γ_1 and Γ_2 are commensurable in G , [dlH00, II.28].*

We follow the terminology of [dlH00], but the reader should beware that groups such that in (i) are also commonly called **abstractly commensurable**, while groups like in (ii) are commonly called just **commensurable**. In general it matters because these are different notions.

Rational forms, lattices, and nilmanifolds.

Theorem A.6 (Mal'cev 1949). *Let G be a real nilpotent, simply connected Lie group, with Lie algebra $\mathfrak{g}$. There are bijections between*

- (i) *G -nilmanifolds, up to homeomorphism and finite covering of the same nilmanifold.*
- (ii) *Lattices of G , modulo the commensurability relation.*
- (iii) *Rational forms of $\mathfrak{g}$, i.e., isomorphism classes of $\mathbf{Q}$ -Lie algebras $\mathfrak{g}_{\mathbf{Q}}$ such that $\mathfrak{g} \simeq \mathfrak{g}_{\mathbf{Q}} \otimes \mathbf{R}$.*

In particular, for a simply connected Lie group G to have a lattice, it is necessary and sufficient that its Lie algebra $\mathfrak{g}$ admits a $\mathbf{R}$ -basis in which the structure constants are rational.

Let us work out the case of the real Heisenberg group. The algebra $\mathfrak{heis}$ is the unique non-abelian nilpotent Lie algebra of dimension 3, regardless of the field of scalars. The Heisenberg group thus admits a unique class of lattices in the sense of (ii). All Heisenberg nilmanifolds of dimension 3 cover

$$V_1 = \text{Heis}_{\mathbf{R}}(3)/\text{Heis}_{\mathbf{Z}}(3);$$

these are torus bundles over the circle S^1 , with parabolic transition matrix $\begin{pmatrix} 1 & \pm n \\ 0 & 1 \end{pmatrix}$ if n is the degree of the covering $V_n \rightarrow V_1$. (More generally, every nilmanifold is the total space of iterated torus bundles.)

Remark A.7 (*). Bridson, de la Harpe, and Kleptsyn show that, once equipped with the Chabauty topology, the space of lattices $\mathcal{R}(\text{Heis}_{\mathbf{R}}(3))$ is a 6-dimensional manifold; it is itself a countable union (indexed by the integer $n \geq 1$) of torus bundles over $(S^3 - T) \times \mathbf{R}$, where T is a trefoil knot in S^3 [BdlHK09] (the factor $S^3 - T$ in the base is $\text{SL}_2(\mathbf{R})/\text{SL}_2(\mathbf{Z})$, the space of lattices of covolume 1 in the plane $\mathbf{R}^2 \simeq \text{Heis}_{\mathbf{R}}(3)^{\text{ab}}$, see [BPS07, p.80]. $\mathcal{R}(\text{Heis}_{\mathbf{R}}(3))$ is a dense open set⁷ in the Chabauty space of the real Heisenberg group.

Example A.8 (Nilpotent Lie group without a lattice). There exist real nilpotent Lie algebras without a rational form. In fact, this is the case for most of them in high dimensions. Here is an argument for this [Rag72, 2.14] in degree 2, though it adapts to higher degrees, see [GMLIP23].

Let E (resp. V) be a real vector space of dimension n (resp. 4). The nilpotent Lie algebra structures $\mathfrak{g}$ of degree 2 on $V \oplus E$ such that $[\mathfrak{g}, \mathfrak{g}] = V$ are parameterized by the set S of surjective⁸ $\alpha \in \Lambda^2 E \otimes V$. S is an open subset of $\Lambda^2 E \otimes V$, which has dimension $2n(n-1)$. The group $\text{GL}(E) \times \text{GL}(V)$ acts on S by

$$[(\varphi, \psi).s](u \wedge v) = \psi(s[\varphi^{-1}(u) \wedge \varphi^{-1}(v)]).$$

⁷A general fact in a nilpotent Lie group [BdlHK09, 3.4(iii)].

⁸ α defines a bracket by setting $[u, v] = \alpha(\bar{u} \wedge \bar{v})$ and $[u, [v, w]] = 0$ for all $u, v, w \in V \oplus E$, where $\bar{x}$ denotes the projection of x onto E parallel to V .

The orbits have dimension at most $n^2 + 16$; however, once bases for V and E are fixed, $\mathfrak{g}$ admits a rational structure if and only if its orbit contains an element with rational coordinates. By the Baire category theorem, as soon as $\dim S > n^2 + 16$, i.e., as soon as $n \geq 6$, these orbits cannot cover S .

Remark A.9. In dimension ≤ 5 , all real nilpotent Lie algebras admit at least one rational form [Cor16, p. 7].

Pfaffian. Theorem A.6 reduces, in principle, the classification of nilmanifolds covered by G , or of lattices in G , to algebra. However, this does not exhaust all the difficulty when it comes to obtaining an explicit classification. We will only treat a few examples in small dimensions. In this sense, it is useful to know how to associate invariants to isomorphism classes of rational Lie algebras, and then to determine which ones are realizable by the rational forms of a given real Lie algebra. The following construction comes from [Lau08]. Let $\mathfrak{g}$ be a nilpotent Lie algebra of degree 2 over the field R (in the following, $R = \mathbf{Q}$ or $\mathbf{R}$). We denote by ad^* its **coadjoint representation**, meaning

$$\langle \text{ad}_X^* Y, Z \rangle = \langle Y, -\text{ad}_X Z \rangle.$$

For every $X \in \mathfrak{g}$, ad_X^* is zero when restricted to $[\mathfrak{g}, \mathfrak{g}]^\perp$, so that $\varphi \mapsto \{(X, Y) \mapsto \langle \text{ad}_X^* \varphi, Y \rangle\}$ induces an endomorphism of $\mathfrak{g}^*/[\mathfrak{g}, \mathfrak{g}]^\perp \simeq [\mathfrak{g}, \mathfrak{g}]^*$ to $\mathfrak{g}^* \wedge \mathfrak{g}^* = \Lambda^2 \mathfrak{g}$, which we denote by J . In fact, $J\varphi$ vanishes on $[\mathfrak{g}, \mathfrak{g}] \times [\mathfrak{g}, \mathfrak{g}]$ for every $\varphi \in [\mathfrak{g}, \mathfrak{g}]^*$ since $\mathfrak{g}$ is of degree 2, and thus J induces $\bar{J} \in \text{Hom}([\mathfrak{g}, \mathfrak{g}]^*, \Lambda^2 \mathfrak{g}_1)$.

Definition A.10 (Pfaffian). Assume $\mathfrak{g}_1$ has even dimension $2m$, and denote by $\mathcal{A}$ the K -algebra of polynomial functions of degree $2m$ on $[\mathfrak{g}, \mathfrak{g}]^*$. Let ω be a form of maximal degree on $\mathfrak{g}_1$. There exists a unique polynomial f on $[\mathfrak{g}, \mathfrak{g}]^*$ such that

$$\forall \varphi \in [\mathfrak{g}, \mathfrak{g}]^*, \quad \Lambda_k \bar{J} \varphi = f(\varphi) \omega.$$

Moreover, according to the Pfaffian theory, there exists $\varepsilon \in \{\pm 1\}$ such that εf is a square in $\mathcal{A}$. Let L be an extension of K . We denote by $\text{Pf}_L(\mathfrak{g})$ the equivalence class in A of a square root of εf under the action of $K^* \times \text{GL}_L([\mathfrak{g}, \mathfrak{g}]^*)$ given by

$$\forall P \in \mathcal{A}, (\lambda, g).P = \lambda P \circ g^{-1}.$$

We call it (or a representative) the **Pfaffian**.

Example A.11. Let $\mathfrak{g} = \mathfrak{heis}_{2m+1}$ be the Heisenberg algebra of dimension $2m + 1$ over K , formed on $(X_1, \dots, X_m, Y_1, \dots, Y_m, Z)$ with $[X_i, Y_j] = \delta_{ij}Z$. Then

$$\bar{J}[tZ^*](X_i, Y_j) = t\delta_{ij},$$

whence $\text{Pf}_K(\mathfrak{g}) = [t^m]$.

The Pfaffian over the field of scalars is an invariant of the isomorphism class of $\mathfrak{g}$. Moreover, the Pfaffian behaves well under an extension of the ground field: if $\mathfrak{g}_{\mathbf{Q}}$ is a rational form of $\mathfrak{g}$, then $\text{Pf}_{\mathbf{R}}(\mathfrak{g}) = \text{Pf}_{\mathbf{R}}(\mathfrak{g}_{\mathbf{Q}})$. Since the classification of forms (for example, quadratic forms) over $\mathbf{Q}$ is considerably finer than that over $\mathbf{R}$ (and a fortiori over $\mathbf{C}$), this does not a priori prevent the existence of a very large number of distinct rational forms for a given real or complex Lie algebra.

An example: Lie algebras of type $(4, 2)$. Still following [Lau08], say that a Lie algebra $\mathfrak{g}$ is of type $(4, 2)$ if it is nilpotent of degree 2, and such that $\dim \mathfrak{g}_1 = 4$ and $\dim[\mathfrak{g}, \mathfrak{g}] = 2$. In this case, the Pfaffian is a class of binary quadratic forms, and it is a complete invariant, by the following theorem.

Proposition A.12 (Grunewald-Segal-Sterling [GSS82]). *Let $\mathfrak{g}, \mathfrak{g}'$ be two Lie algebras of type $(4, 2)$ over a field of characteristic zero, with the same Pfaffian. Then $\mathfrak{g}$ and $\mathfrak{g}'$ are isomorphic.*

Since binary quadratic forms are themselves classified by their discriminant, the previous proposition gives the classification of Lie algebras of type $(4, 2)$

(a) over the rationals: The rational Lie algebras of type $(4, 2)$ are indexed by $\mathbf{Q}/\mathbf{Q}^{\times 2}$ and a complete system of representatives is given by

$$\mathfrak{n}_k := \langle X_1, \dots, X_4, Z_1, Z_2 \rangle$$

with the Lie brackets

$$\begin{aligned} [X_1, X_3] &= [X_2, X_4] = Z_1, [X_1, X_4] = Z_2, \\ [X_1, X_2] &= [X_3, X_4] = 0, [X_2, X_3] = kZ_2. \end{aligned}$$

for every integer k , zero or square-free. We verify using the previous basis that

$$\frac{\Lambda_2 J(xZ_1^* + yZ_2^*)}{X_1^{\text{ab}} \wedge \dots \wedge X_4^{\text{ab}}} = \det \begin{pmatrix} 0 & 0 & x & y \\ 0 & 0 & ky & x \\ -x & -ky & 0 & 0 \\ -y & -x & 0 & 0 \end{pmatrix} = (x^2 - ky^2)^2.$$

Hence $\text{Pf}_{\mathbf{Q}}(\mathfrak{n}_k) \equiv x^2 - ky^2$, with discriminant k .

(b) over the reals: The discriminant retains only the sign, and there are 3 isomorphism classes of Lie algebras of type $(4, 2)$: for every $k > 0$

$$\begin{aligned} \mathfrak{n}_- &\simeq \mathfrak{n}_k \otimes \mathbf{R} & k < 0 \\ \mathfrak{n}_0 &\simeq \mathfrak{n}_0 \otimes \mathbf{R} \\ \mathfrak{n}_+ &\simeq \mathfrak{n}_k \otimes \mathbf{R} & k > 0 \end{aligned}$$

We verify that $\mathfrak{n}_1 \otimes \mathbf{R} = \mathfrak{heis} \oplus \mathfrak{heis}$ by setting $Y_1^\pm = (X_1 \pm X_2)/2$, $Y_2^\pm = (X_3 \pm X_4)/2$ and $Z^\pm = (Z_1 \pm Z_2)/2$, and by calculating $[X^\pm, Y^\pm] = Z^\pm$ and $[X^\pm, Y^\mp] = 0$. Similarly, $\mathfrak{n}_{-1} \otimes \mathbf{R}$ is the Lie algebra underlying $\mathfrak{heis}(\mathbf{C})$.

Remark A.13. We have given examples with zero, one, or infinitely many rational forms. According to [dlH00, p. 304], the question of whether there exists a real nilpotent Lie algebra with a finite number $k \neq 1$ of rational forms is open.

Cohomology rings of nilmanifolds and groups. Following the theorems of Mal'cev on nilmanifolds, Matsushima (in degrees 1 and 2) and Nomizu were able to prove the following.

Theorem A.14 ([Mat51, Nom54]). *Let Γ be a finitely generated nilpotent group and let $\mathfrak{g}$ be the Lie algebra of the real Mal'cev completion of Γ . Then $H^*(\Gamma, \mathbf{R}) \simeq H^*(\mathfrak{g}, \mathbf{R})$*

Remark A.15 (Historical comment). it is after hearing a talk by Nomizu on this theorem that Armand Borel stated his well-known conjecture in a letter to Serre in May 1953, see [a copy of the letter on Andrew Ranicki's website](#).

In relation to the open problem of this lecture, Nomizu's theorem is now subsumed by the following modern statement where the real cohomology rings appear to be quasiisometric invariants.

Theorem A.16 ([Sha04, Sau06, GK21, AY25]). *Let G and H be either finitely generated nilpotent groups, or simply connected nilpotent Lie groups. If G and H are quasiisometric, then their real cohomology rings $H^*(G, \mathbf{R})$ and $H^*(H, \mathbf{R})$ are isomorphic.*

APPENDIX B. SOLUTIONS OF THE EXERCISES

Solution of Exercise A. By definition, for every $X, Y, Z \in \mathfrak{g}$ we have

$$\mathrm{ad}_{[X,Y]}(Z) = [[X, Y], Z] = [X, [Y, Z]] + [Y, [Z, X]]$$

where we used (J) and antisymmetry. On the other hand, the Lie bracket in $\mathfrak{gl}(\mathfrak{g})$ is given by $[f, g] = fg - gf$, especially

$$[\mathrm{ad}_X, \mathrm{ad}_Y](Z) = [X, [Y, Z]] - [Y, [X, Z]] = [X, [Y, Z]] + [Y, [Z, X]].$$

This shows that $\mathrm{ad}_{[X,Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y]$ as required.

Solution of Exercise B.

- (1) Let $\mathfrak{h}$ be a three-dimensional nilpotent, non-abelian Lie algebra. Let X and Y in $\mathfrak{h}$ be such that $Z := [X, Y]$ is nonzero. X and Y are linearly independent, otherwise their bracket would vanish. Assume towards a contradiction that $Z = \alpha X + \beta Y$ with $\beta \neq 0$. Then $[X, Z] = \beta Z \neq 0$. But then $C^3\mathfrak{h} = C^2\mathfrak{h} = \mathrm{span}(Z)$, and the central series of $\mathfrak{h}$ does not stabilize at zero. A similar argument shows that α must be zero as well. So (X, Y, Z) span a basis of $\mathfrak{h}$, and $\mathfrak{h}$ is isomorphic to $\mathfrak{heis}$.
- (2) Let $\mathfrak{g}$ be a 4-dimensional nilpotent Lie algebra. We will distinguish cases along the nilpotency class of $\mathfrak{g}$.

Case 1: $\mathfrak{g}$ is 1-nilpotent. Then it is abelian, and isomorphic to $\mathbf{R}^4$.

Case 2: $\mathfrak{g}$ is 2-nilpotent. In the beginning of the proof of Proposition and Definition 1.7 we showed that $\dim \mathfrak{g}/C^2\mathfrak{g} > 1$. This still applies in our case, and we know in addition that $\dim C^2\mathfrak{g} \geq 1$, so that $\mathfrak{g}/C^2\mathfrak{g}$ has dimension 2 or 3. Let φ be a linear form on $\mathfrak{g}$ such that $C^2\mathfrak{g} \cap \ker \varphi = 0$ and consider the bilinear form $\lambda: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbf{R}$, $\lambda(X, Y) = \varphi([X, Y])$. We know that $C^3\mathfrak{g} = 0$; and by Lemma 1.4, for all $Z \in C^2\mathfrak{g}$ and $X \in \mathfrak{g}$, $[X, Z] \in C^3\mathfrak{g}$, so $\lambda(X, Z) = 0$. It follows that λ defines yet another antisymmetric, bilinear form $\underline{\lambda}$ on $\mathfrak{g}/C^2\mathfrak{g}$; it is nonzero since $C^2\mathfrak{g} \cap \ker \varphi = 0$. By the reduction theory of antisymmetric bilinear forms on a two or three-dimensional vector space, there is a basis $(\underline{X}, \underline{Y})$ of $\mathfrak{g}/C^2\mathfrak{g}$ (if $\dim \mathfrak{g}/C^2\mathfrak{g} = 2$) or $(\underline{X}, \underline{Y}, \underline{W})$ (if $\dim \mathfrak{g}/C^2\mathfrak{g} = 3$) in which $\underline{\lambda}$ has matrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

respectively. Now lift $\underline{X}, \underline{Y}$ and (if necessary) $\underline{W}$ to X, Y and (if necessary) W in $\mathfrak{g}$ and let $Z = [X, Y] \in C^2\mathfrak{g} \setminus \{0\}$. We have that $C^2\mathfrak{g} = \mathrm{span}(Z)$, which shows that $\dim C^2\mathfrak{g} = 1$ and excludes the case

where $\dim C^2\mathfrak{g} = 2$. The vectors (X, Y, W, Z) form a basis of $\mathfrak{g}$ and the only nonzero bracket among them is $Z = [X, Y] = -[Y, X]$. Thus $\mathfrak{g} \simeq \mathfrak{heis} \times \mathbf{R}$.

Case 3: $\mathfrak{g}$ is 3-nilpotent. By Proposition 1.7 we know that $\mathfrak{g}$ is filiform. In particular, $\dim C^3\mathfrak{g} = 1$. Then $\mathfrak{h} := \mathfrak{g}/C^3\mathfrak{g}$ is a non-abelian, three-dimensional nilpotent Lie algebra, hence it is $\mathfrak{h} = \mathfrak{heis}$ by the result of Question 1. Thus there exists a basis (X_0, Y_0, Z_0, W_0) of $\mathfrak{g}$ such that $[X_0, Y_0] = Z_0$, $[X_0, Z_0] = W_0$, $[Y_0, Z_0] = \alpha W_0$ for some $\alpha \in \mathbf{R}$, and $[X_0, W_0] = [Y_0, W_0] = [Z_0, W_0] = 0$. Letting $X = X_0$, $Y = Y_0 - \alpha Z_0$, $Z = Z_0$ and $W = W_0$ we get that $\mathfrak{g} \simeq \mathfrak{l}_4$.

If $\dim C^2\mathfrak{g} = 2$, let X and Y span a supplementary subspace to $C^2\mathfrak{g}$ in $\mathfrak{g}$.

Solution of Exercise C.

- (1) Since $\mathfrak{h}$ is 2-nilpotent, we have that $0 \neq C^2\mathfrak{h} \subseteq Z(\mathfrak{h})$; since additionally $\dim Z(\mathfrak{h}) = 1$ we have that

$$C^2\mathfrak{h} = Z(\mathfrak{h}) = \text{span}(Z)$$

for some $Z \in \mathfrak{h}$. Then there is a bilinear form λ on $\mathfrak{h}/Z(\mathfrak{h})$ such that $[X, Y] = \lambda(X + Z(\mathfrak{h}), Y + Z(\mathfrak{h}))Z$ for all $X, Y \in \mathfrak{h}/Z(\mathfrak{h})$. This bilinear form is non-degenerate and antisymmetric, thus $\dim \mathfrak{h}/C^2\mathfrak{h}$ is even-dimensional. (Note that we already used a variant of this argument in Question 2 of Exercise B.)

- (2) A non-degenerate antisymmetric bilinear form on a $2n$ -dimensional vector space can always be written in the form $e_1^* \wedge f_1^* + \dots + e_n^* \wedge f_n^*$ where $e_1, \dots, e_n, f_1, \dots, f_n$ is a basis. Applying this to λ as in Question 1 and letting $X_i = e_i$ and $Y_j = f_j$ for $i, j = 1, \dots, n$ we get the conclusion.
- (3) Consider $V = \text{span}(\{E_{1,i}\}_{2 \leq i \leq n+1} \cup \{E_{j,n+2}\}_{2 \leq j \leq n+1} \cup \{E_{n+2,n+2}\})$. This is the vector space of strictly upper triangular matrices which are allowed to have nonzero entries only on the top row and rightmost column. A computation gives that

$$[E_{1,i}, E_{j,n+2}] = E_{1,i}E_{j,n+2} - E_{j,n+2}E_{1,i} = \mathbf{1}_{i=j}E_{n+2,n+2}$$

for all $i, j = 1, \dots, n$. Letting $X_i = E_{1,i}$ and $Y_j = E_{j,n+2}$ we see that this is a Lie subalgebra of $\mathfrak{u}_{n+2}$ isomorphic to $\mathfrak{h}$.

Solution of Exercise D.

- (1) $C^2\mathfrak{l}'_5 = \text{span}(X_3, X_4, X_5)$, $C^3\mathfrak{l}'_5 = \text{span}(X_4, X_5)$ and $C^4\mathfrak{l}'_5 = \text{span}(X_5)$. Since $[X_2, X_3] = X_5 = 0 \pmod{C^3\mathfrak{l}'_5}$ and all the other brackets are equal to those in the Carnot Lie algebra $\mathfrak{l}_5$, we have that $(\mathfrak{l}'_5)_\infty = \mathfrak{l}_5$.
- (2) Given $\mathfrak{l} \in \{\mathfrak{l}_5, \mathfrak{l}'_5\}$ consider the integer

$$m(\mathfrak{l}) = \inf \{ \text{rank}(\text{ad}_X) : X \in \mathfrak{l} \setminus C^2\mathfrak{l} \}.$$

In both cases the center is contained in $C^4\mathfrak{l}$, so $m(\mathfrak{l}) \geq 1$. In $\mathfrak{l}_5$, $\text{rank}(\text{ad}_{X_2}) = \dim \text{span}(X_3) = 1$, so $m(\mathfrak{l}_5) = 1$. Consider now $X = \alpha_1 X_1 + \dots + \alpha_5 X_5$ in $\mathfrak{l}'_5$, and assume that α_1 and α_2 are not both zero so that $X \notin C^2\mathfrak{l}'_5$. A computation gives that $[X, X_3] = \alpha_1 X_4 + (\alpha_1 + \alpha_2)X_5$ while $[X, X_4] = \alpha_1 X_5$. These two vectors are linearly independent, so if $\alpha_1 \neq 0$ then ad_X has rank at least 2. Assuming now that $\alpha_1 = 0$ we get that $[X, X_5] = \alpha_2 X_5$ while $[X, X_1] = -\alpha_2 X_3$. Again, these two vectors are linearly independent, which proves that $\text{rank}(\text{ad}_X) \geq 2$ for all $X \in \mathfrak{l}'_5 \setminus C^2\mathfrak{l}'_5$. Thus $m(\mathfrak{l}'_5) \geq 2$ (and we can in

fact check that $m(\mathfrak{l}'_5) = 2$). But m is invariant under Lie algebra isomorphisms. So $\mathfrak{l}_5$ and $\mathfrak{l}'_5$ are not isomorphic.

Solution of Exercise E.

- (1) Let V_1 be a supplementary subspace to $C^2\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. Then $[V_1, V_1] = C^2\mathfrak{g}$. Setting $\mathfrak{g}_1 = V_1$ and $\mathfrak{g}_2 = C^2\mathfrak{g}$ we have that $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ is a Carnot grading.
- (2) Consider the Lie algebra isomorphisms $\gamma: \mathfrak{g} \rightarrow \mathfrak{g}_\infty$ and $\gamma': \mathfrak{g} \rightarrow \mathfrak{g}_\infty$ mapping $\mathfrak{g}_i$ (resp. $\mathfrak{g}'_i$) to $(\mathfrak{g}_\infty)_i$ and consider $\gamma^{-1}\gamma'$. This is an automorphism of $\mathfrak{g}$ sending $\mathfrak{g}'_i$ onto $\mathfrak{g}_i$ for all i .

Solution of Exercise F.

- (1) Let s be the nilpotency class of $\mathfrak{g}$. Let $i, j \in \{1, \dots, s\}$, $X \in \mathfrak{g}_i$ and $Y \in \mathfrak{g}_j$. Then $[X, Y] \in \mathfrak{g}_{i+j}$, and

$$-[X, [Y, T]] - [Y, [T, X]] = [X, jY] - [Y, iX] = (j+i)[X, Y] = [T, [X, Y]].$$

Since T together with the $\mathfrak{g}_i$ linearly span $\mathfrak{h}$, this shows that the Jacobi identity holds in $\mathfrak{h}$, and so $\mathfrak{h}$ is a Lie algebra indeed. (There is a more general construction called a semidirect product of Lie algebras. Here $\mathfrak{h} = \mathfrak{g} \rtimes \mathbf{R}$.)

- (2) We know from Exercise A that ad is a homomorphism of Lie algebras. Its kernel is equal to $Z(\mathfrak{h})$, which is trivial by Question 1. Restricting to the Lie subalgebra $\mathfrak{g}$ of $\mathfrak{h}$, we get that $\mathfrak{g}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}_{\dim \mathfrak{g}+1}(\mathbf{R})$.
- (3) Set $e_{n+1} = T$ and then take a basis of $\mathfrak{g}$ by concatenating bases of $\mathfrak{g}_s, \dots, \mathfrak{g}_1$ (in this order) to form $e_1, \dots, e_n$. In this way we have $[e_i, e_j] \in \text{span}\{e_k : k < i\}$ for all $i < j$, and $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{u}_{n+1}$. (One may note that $\mathfrak{h}$ is still graded by setting $\mathfrak{h}_0 = \text{span}(T)$ and $\mathfrak{h}_i = \mathfrak{g}_i$ for $i > 0$; the basis is produced by concatenating bases of $\mathfrak{h}_s, \dots, \mathfrak{h}_0$ in this order.)

Beware: A mistake (that might be produced by mixing the statements of two theorems of Engel and Lie) is to believe that nilpotent Lie subalgebras of $\mathfrak{gl}_n(\mathbf{C})$ can always be conjugated into a Lie subalgebra of $\mathfrak{u}_n$. A counter-example is given by a Lie subalgebra of diagonal matrices, which is abelian, hence nilpotent.

- (4) First, it is indeed an extra case because, as we showed in Exercise D, $\mathfrak{l}'_5$ is not isomorphic to its associated Carnot-graded Lie algebra, hence (by Proposition and Definition 2.1) it is not a Carnot Lie algebra. However, note that $\mathfrak{l}'_5$ has a basis $(X_1, \dots, X_5)$ such that $[X_i, X_j]$ is either 0 or $\pm X_{i+j}$ for every $i, j = 1, \dots, 5$. Hence, setting $V_i = \text{span}(X_i)$ we get that $V_1 \oplus \dots \oplus V_5$ is a Lie algebra grading valued in the positive integers. Examining Question 1, we see that it is sufficient in order to define a center-less, larger Lie algebra $\mathfrak{h}$ with an element T that normalizes $\mathfrak{g}$ and such that $[T, X_i] = iX_i$ for $i = 1, \dots, 5$. There we did not use the fact that the grading was Carnot; the crucial fact was that $\mathfrak{g}_0 = 0$. After that, the reasoning given for Questions 2 and 3 applies.

Solution of Exercise G.

- (1) Some matrix computations give, for every x, y, z, p, q , and r in $\mathbf{R}$,

$$\exp \begin{pmatrix} 0 & p & r \\ 0 & 0 & q \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & p & r + \frac{pq}{2} \\ 0 & 1 & q \\ 0 & 0 & 1 \end{pmatrix} \text{ and } \log \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & x & z - \frac{xy}{2} \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix}.$$

Now let $M_i = p_i E_{12} + q_i E_{23} + r_i E_{13}$ for $i = 1, 2$; combining the above, we get

$$\begin{aligned}
 \log(\exp(M_1) \exp(M_2)) &= \log \left(\begin{pmatrix} 1 & p_1 & r_1 + \frac{p_1 q_1}{2} \\ 0 & 1 & q_1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & p_2 & r_2 + \frac{p_2 q_2}{2} \\ 0 & 1 & q_2 \\ 0 & 0 & 1 \end{pmatrix} \right) \\
 &= \log \begin{pmatrix} 1 & p_1 + p_2 & r_1 + r_2 + \frac{p_1 q_1 + p_2 q_2}{2} + p_1 q_2 \\ 0 & 1 & q_1 + q_2 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & p_1 + p_2 & r_1 + r_2 + \frac{2p_1 q_2 + p_1 q_1 + p_2 q_2 - (p_1 + p_2)(q_1 + q_2)}{2} \\ 0 & 0 & q_1 + q_2 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & p_1 + p_2 & r_1 + r_2 + \frac{p_1 q_2 - p_2 q_1}{2} \\ 0 & 0 & q_1 + q_2 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

as we wanted.

- (2) We want to use a nilpotent Lie algebra of class 3 to let the triple brackets appear. Let us embed $\mathfrak{l}_4$ (which is the smaller-dimensional of these) into $\mathfrak{gl}_5(\mathbf{R})$ using the method of Exercise F. We will use the basis $(e_1, \dots, e_5) = (X_4, X_3, X_2, X_1, T)$ of $\mathfrak{h} = \mathfrak{l}_4 \oplus \text{span}(T)$. We have

$$\text{ad}_{X_4} = -3T \otimes (X_4) = -3E_{15}$$

$$\text{ad}_{X_3} = -2T \otimes (X_3) - X_1 \otimes X_4 = -2E_{25} - E_{14}$$

$$\text{ad}_{X_2} = -T \otimes (X_2) - X_1 \otimes X_3 = -E_{35} - E_{24}$$

$$\text{ad}_{X_1} = -T \otimes (X_1) + X_2 \otimes X_3 + X_3 \otimes X_4 = -E_{45} + E_{23} + E_{12}$$

In this way, for every $p, q, r, s \in \mathbf{R}$, the element $pX_1 + qX_2 + rX_3 + sX_4$ is represented by

$$M(p, q, r, s) = \begin{pmatrix} 0 & p & 0 & -r & -3s \\ 0 & 0 & p & -q & -2r \\ 0 & 0 & 0 & 0 & -q \\ 0 & 0 & 0 & 0 & -p \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

We now need to perform a few 5×5 matrix computations. For this we use a little help from [the WIMS matrix calculator available here](#). We get

$$\exp M(p, q, r, s) = \begin{pmatrix} 1 & p & \frac{p^2}{2} & -\frac{pq}{2} - r & -\frac{pq}{2} - 3s \\ 0 & 1 & p & -q & -2r \\ 0 & 0 & 1 & 0 & -q \\ 0 & 0 & 0 & 1 & -p \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

It remains to compute $\log(\exp M(p_1, \dots, s_1) \exp M(p_2, \dots, s_2))$. We find the product of matrix exponentials to equal

$$\begin{pmatrix} 1 & p_1 + p_2 & p_1^2/2 + p_1 p_2 + p_2^2/2 & (-q_1/2 - q_2)p_1 - 2q_1 - p_2 q_2/2 & f(p_1, q_1, \dots, r_2, s_2) \\ 0 & 1 & p_1 + p_2 & (-q_1 - q_2) & -p_1 q_2 + (p_2 q_1 - 2(q_1 + q_2)) \\ 0 & 0 & 1 & 0 & -q_1 - q_2 \\ 0 & 0 & 0 & 1 & -p_1 - p_2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

where

$$f(p_1, q_1, \dots, r_2, s_2) = -\frac{p_1^2 q_2}{2} + p_1 \left(\frac{p_2 q_1}{2} + \frac{q_1}{2} - 2q_2 \right) + p_2 q_1 - 3s_1 - \frac{p_2 q_2}{2} - 3s_2$$

and then $\log(\exp M(p_1, \dots, s_1) \exp M(p_2, \dots, s_2))$ is

$$\begin{pmatrix} 0 & p_1 + p_2 & 0 & \frac{p_2 q_1 - p_1 q_2}{2} - 2r_1 & g(p_1, \dots, s_2) \\ 0 & 0 & p_1 + p_2 & -q_1 - q_2 & p_2 q_1 - p_1 q_2 - 2(r_1 + r_2) \\ 0 & 0 & 0 & 0 & -q_1 - q_2 \\ 0 & 0 & 0 & 0 & -p_1 - p_2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where $g(p_1, \dots, s_2) = -\frac{p_1^2 q_2 + p_2^2 q_1}{4} + g_2(p_1, \dots, s_2)$. Here in g_2 we gather the terms of homogeneous polynomial degree at most 2, we are not really interested in looking at them in relation with the third term of the BCH series. (The terms just above the diagonal are consistent with computations in $\mathfrak{l}_4$ modded out by the first two rms in its lower central series and can be used as a sanity check for our computations.) Inspecting g we find that 3 times the coefficient in front of $X_4 = [X_1, [X_1, X_2]]$ coming out of the BCH expansion of $\log(\exp M_1 \exp M_2)$ must equal $\frac{1}{4}(p_2^2 q_1 - p_1^2 q_2)$. It implies that the coefficients in front of the iterated brackets $[p_1 X_1 + q_1 X_2, [p_1 X_1 + q_1 X_2, p_2 X_2 + q_2 X_2]]$ and $[p_2 X_1 + q_2 X_2, [p_2 X_1 + q_2 X_2, p_1 X_2 + q_1 X_2]]$ must be $\pm \frac{1}{3} \times \frac{1}{4} = \pm \frac{1}{12}$.

Solution of Exercise H.

We will denote by $\gamma(G)$ the order of polynomial growth of a nilpotent group G .

- (1) For $G = \mathbf{Heis}$, we have that $\dim C^2 \mathfrak{g} = 1$, hence $\gamma(G) = 2 + 2 = 4$. Similarly, $\dim C^2 \mathfrak{l}_4 = 2$ and $\dim C^3 \mathfrak{l}_4 = 1$ so $\gamma(L_4) = 2 + 2 + 3 = 7$. Finally $\gamma(\mathbf{Heis} \times \mathbf{R}) = 4 + 1 = 5$.
- (2) Γ has its growth bounded by a polynomial, hence by Gromov's polynomial growth theorem it is virtually nilpotent. Let N be Mal'cev completion of a torsion free finitely generated nilpotent subgroup of Γ (This is also the connected nilshadow of Γ from Theorem 0.1. Since $\pi < 4$, we know that $\dim G < 4$. So G is either 1, $\mathbf{R}$, $\mathbf{R}^2$, $\mathbf{R}^3$ or $\mathbf{Heis}$. But in Question 1 we computed that $\gamma(\mathbf{Heis}) = 4$. So N is either 1, $\mathbf{R}$, $\mathbf{R}^2$ or $\mathbf{R}^3$, and Γ being virtually a lattice in N is virtually free abelian.
- (3) Since growth is a quasiisometric invariant, we know that $\gamma(\Gamma)$ is 1, 2 or 3. Reproducing the same reasoning as before we see that the connected nilshadow of Γ is either $\mathbf{R}$, $\mathbf{R}^2$ or $\mathbf{R}^3$. Then Γ is virtually either $\mathbf{Z}$, $\mathbf{Z}^2$ or $\mathbf{Z}^3$.

Solution of Exercise J.

- (1) By Theorem 5.17, having polynomially bounded growth is a SBE invariant. Since Γ is $O(r^e)$ -SBE to $\mathbf{Z}^d$, it has polynomial growth. Now, let G be the connected nilshadow of Γ . We know that G and $\mathbf{R}^d$ are $O(r^e)$ -SBE. By Pansu's theorem 5.12, $G_\infty \simeq \mathbf{R}^d$. It follows that $G = \mathbf{R}^d$, and Γ is virtually $\mathbf{Z}^d$.

Note: *a posteriori*, the exponent of polynomial growth (when the growth is polynomially bounded) is a SBE invariant.

- (2) First, let us make sense of the statement. It is not a priori completely clear why the group $\Lambda = \mathbf{Heis}(\mathbf{Z}[\sqrt{2}])$ is indeed finitely generated and this why one

can speak of its QI type. Let

$$g = \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \text{ with } x, y, z \in \mathbf{Z}[\sqrt{2}].$$

Then we may decompose g into a product

$$g = \begin{pmatrix} 1 & x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & z - xy \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and now x, y and $\zeta = z - xy$ are all in $\mathbf{Z}[\sqrt{2}]$, we may write the three matrices as products of elements in the set

$$\begin{pmatrix} 1 & \pm 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & \pm\sqrt{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \pm 1 \\ 0 & 0 & 1 \end{pmatrix}, \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \pm\sqrt{2} \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \pm 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & \pm\sqrt{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

of cardinal 12 (this is not a minimal generating set). From now on we equip Λ with the word distance of this generating set (and *not* of the distance it inherits as a dense subgroup of the Heisenberg group).

We may embed Λ in $L = \mathbf{Heis}(\mathbf{R}) \times \mathbf{Heis}(\mathbf{R})$ by sending $g \mapsto (g, \bar{g})$, where $\bar{g}$ has the conjugate coordinates to those of g . We can check that it is a lattice there, either by invoking the general theorem of Borel–Harisch–Chandra or using Theorem A.2 and checking that the image is Zariski-dense.

If Γ is a group quasiisometric to Λ , then Γ has polynomial growth (of order 8); let G be its connected nilshadow. By Pansu’s theorem 5.12, $G_\infty \simeq L$. Hence G_∞ has nilpotency class 2. By the result of the first question of Exercise E, G is Carnot-graded, hence $G \simeq L$. So Γ is virtually a lattice in L .

It may be the case that Γ and Λ are not commensurable. Indeed, consider $\Gamma' = \mathbf{Heis}(\mathbf{Z}[\sqrt{3}])$. Then $\log \Gamma'$ is contained in a $\mathbf{Q}$ -Lie algebra isomorphic to $\mathfrak{n}_3$ (in the notation of the appendix) while $\log \mathbf{Heis}(\mathbf{Z}[\sqrt{2}])$ is contained in a Lie algebra isomorphic to $\mathfrak{n}_2$. Since $\mathfrak{n}_2$ and $\mathfrak{n}_3$ are not isomorphic (indeed they are Lie algebras of type $(4, 2)$ with Pfaffian of discriminant 2 and 3 respectively), the groups Γ and Γ' are not commensurable (See Theorem A.6). (Another way to state this is that the rational Malcev completions $\Gamma_{\mathbf{Q}} = \mathbf{Heis}(\mathbf{Q}[\sqrt{2}])$ and $\Gamma'_{\mathbf{Q}} = \mathbf{Heis}(\mathbf{Q}[\sqrt{3}])$ are not isomorphic.)

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